EST. ON THE CRISPY FRIED CHICKEN BOARD · V3S · 23 OCTOBER 2026

Dark River Tavern Menu

Every dish on the board, plated with its data: what was measured, what the theory predicted, and why it passes, fails or waits.

PASSES **7** | CONDITIONAL **9** | NOT DECIDABLE **5** | FAILS **10** | OPEN **9**

House rules. 40 dishes, one per board tile, each colour exactly as the board has it. Every number in a picture is read from a committed lane file, named under the dish. $\kappa = ½$ is fitted, not derived; the two footings (canonical 9.3603×10^{-11}, alternative 1.1312×10^{-10} m s^{-2}) are never pooled. Nothing here says the data favour this theory or any rival, and the theory is not closed.

No ΛCDM on the stove. No picture uses ΛCDM as the yardstick. Where a lane leaned on a ΛCDM-derived input, the dish says so.

Which kitchen. This is the 23 October board, scored for v3s. On 25 October you chose a new flagship (the a0 law plus cold dark energy with the core law, PAPER61), and the board has not been redrawn for it.

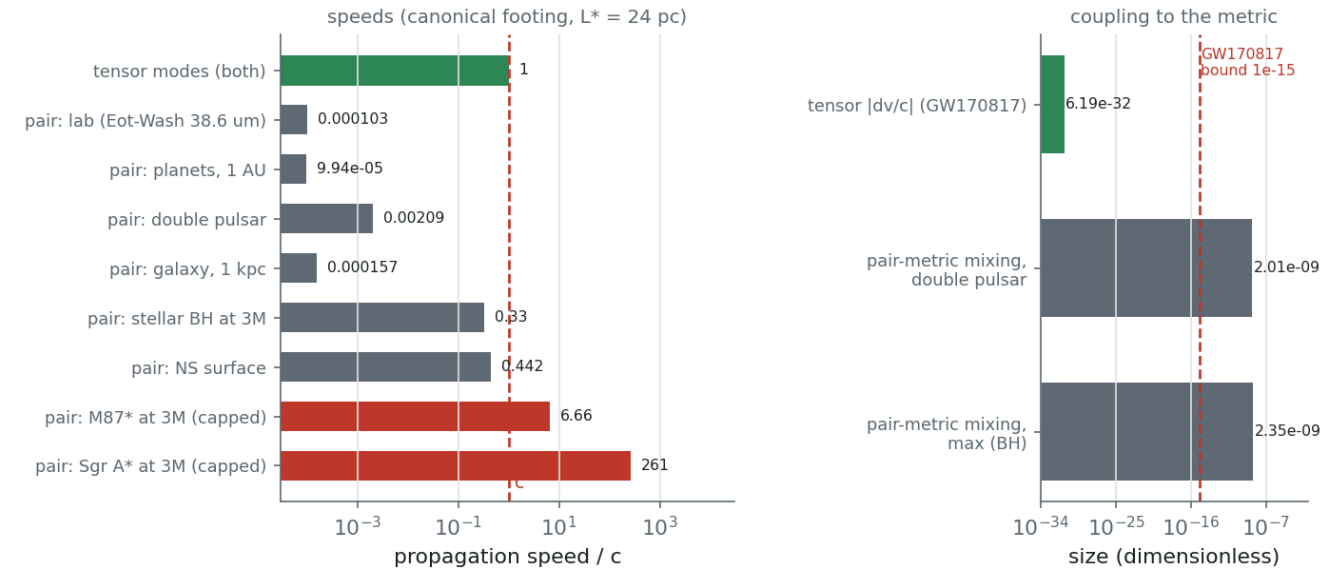
~ Starters ~

The chassis: the relativistic theory underneath · 10 dishes

Gravity waves at light speed

PASSES

On the board: tensor modes at c exactly; where D·P > 0 the khronon/dust pair also propagates (P3: ~1e-4 c in the Solar System), metric coupling ≤ 2.3e-9



source: covariant_push_2026_10_22/C5_health/c5_health_results.json (H4 rows); rebuild_2026_10_22/P3_v3_strong_field/p3_v3_strong_field_results.json (E4, mixing); covariant_push_2026_10_22/C8_instantaneous/c8_instantaneous_results.json (K9)

ON THE PLATE No observational data array: the linear spectrum of v3 (beta = eta = 0, BS dust) from C5, the khronon/dust pair speeds from P3's decoupling-limit estimate on 8 declared backgrounds (canonical footing, L* = 24 pc), the tensor speed shift from C8 K9.

HOW IT IS PLATED Left: horizontal log bars of propagation speed in units of c, with c drawn dashed; the two tensor modes sit exactly on 1, the pair's speed per background in grey, red where it exceeds c (capped supermassive holes). Right: log bars of the tensor |dv/c| at 100 Hz against the GW170817 bound 1e-15, and the pair-to-metric mixing.

WHY IT PASSES With beta = 0 the tensor sector is GR's, so both tensor modes have omega^2/k^2 = 1 exactly on every background tested. Where D.P > 0 the khronon/dust pair also propagates (P3 corrects the earlier 'tensor only' reading): about 1e-4 c in the lab and Solar System, 2.1e-3 c at the double pulsar, but superluminal near capped supermassive holes (6.66 c at M87*, 261 c at Sgr A*). Its mixing with metric waves is at most 2.35e-9, and the induced tensor speed shift is at most 6.19e-32, far below 1e-15. CFG292 (old chassis, also beta = 0) has the same tensor sector.

Sources: covariant_push_2026_10_22/C5_health/c5_health_results.json @ f3e7d4d661; rebuild_2026_10_22/P3_v3_strong_field/p3_v3_strong_field_results.json @ 2b067f2813; covariant_push_2026_10_22/C8_instantaneous/c8_instantaneous_results.json @ aa113a5e96;

campaign_fresh_gravity/CFG292_khronon_strong_hyperbolicity/cfg292_khronon_strong_hyperbolicity_results.json @ 918b331834

Lanes: C5 / C8 / CFG292 / P3

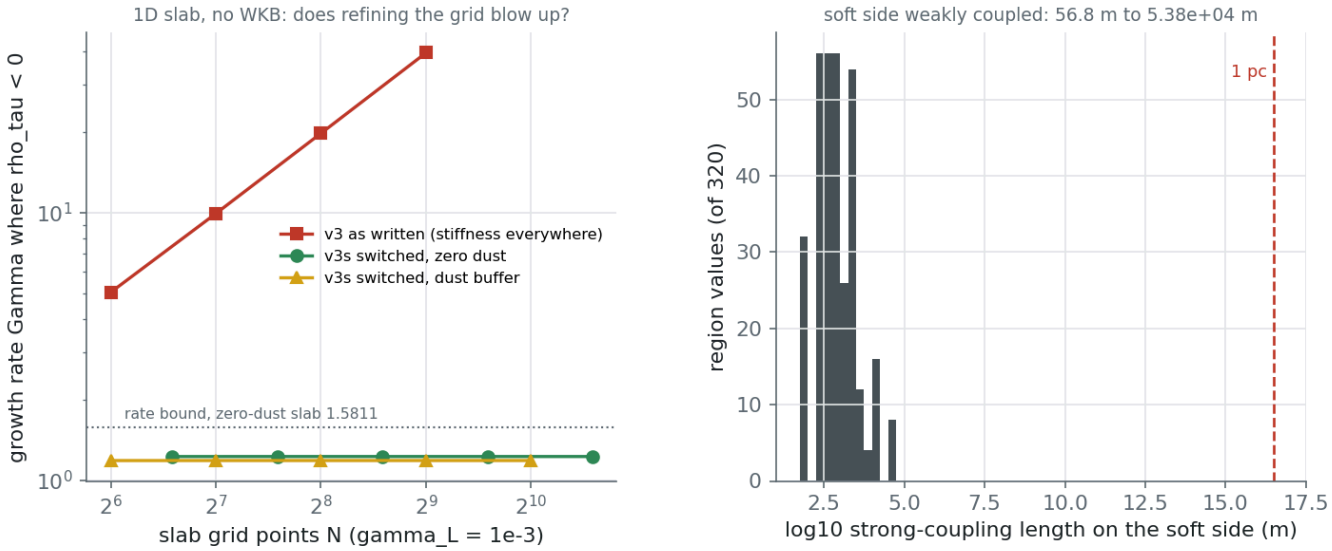
Health: ghosts, gradients

CONDITIONAL

On the board: v3s: switched stiffness well-posed where $\rho_\tau < 0$ (X3: 1D nonlinear $c^2 \geq 0$, slab converges; Lean); if dust ≥ −D·P; zero-dust edge; soft side weakly coupled (≥ 57 m); 2D/3D open (v3: H4 unbounded)

Health: ghosts, gradients

CONDITIONAL: the switch stops the blow-up in 1D; needs dust >= -D.P, has a zero-dust knife edge; 2D/3D open



source: rebuild_2026_10_22/X3_switched_stiffness/x3_switched_stiffness_results.json (C); reviews/v3s_soft_side_strong_coupling_2026_10_23/soft_side_sc_results.json (N3)

ON THE PLATE

X3's 1D periodic slab (no WKB) growth rate Gamma where rho_tau < 0, for v3 as written and for the v3s switched stiffness; the 320 soft-side strong-coupling lengths from the 10-23 review (H4 region values, both footings). Ghost-freedom of the literal form is C5's linear spectrum.

HOW IT IS PLATED

Left: Gamma versus grid points N on log axes; v3 (red squares) doubles with each doubling of N, the switched operator (green, zero dust; gold, dust buffer) stays flat below the analytic rate bound. Right: histogram of log10 of the 320 soft-side strong-coupling lengths with 1 pc drawn dashed.

WHY IT IS CONDITIONAL

v3 as written is Hadamard ill-posed where rho_tau < 0: the slab rate doubles per grid doubling (5.04 to 39.6), H4's UV-unbounded growth. Switching the stiffness off where D.P < 0 gives a rate that converges (1.224744 with zero dust, 1.183214 with a dust buffer) and stays under the bound, with c^2 >= 0 for every state. It is conditional: it needs dust >= -D.P in stiff regions, the zero-dust interface is a knife edge (a scheme that lets stiffness touch a rho_tau < 0 face grows as N^0.497), and 2D/3D evolution is not computed. On the soft side the strong-coupling length is 56.8 m to 5.38e4 m, far below 1 pc, so the effective theory holds there.

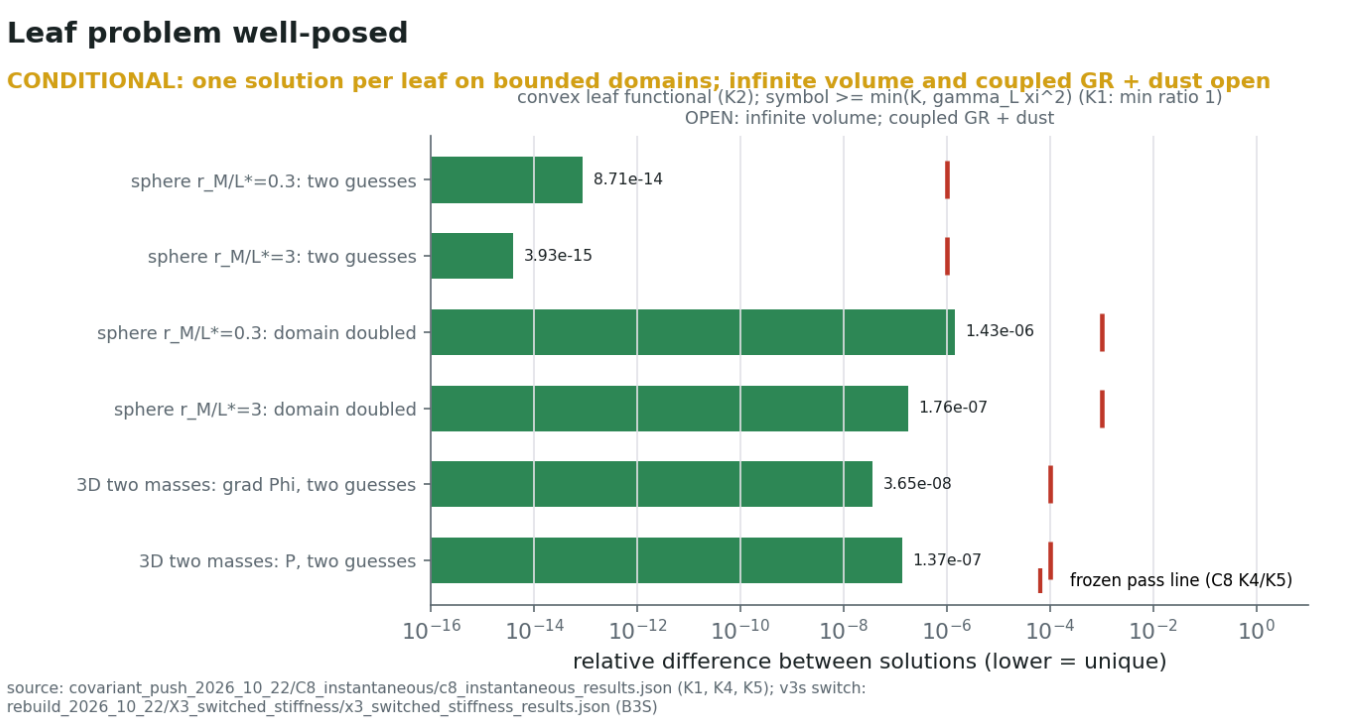
Sources: rebuild_2026_10_22/X3_switched_stiffness/x3_switched_stiffness_results.json @ 31febd3b70; rebuild_2026_10_22/X3_switched_stiffness/README.md @ 9912051760; reviews/v3s_soft_side_strong_coupling_2026_10_23/soft_side_sc_results.json @ d27e01f42e; rebuild_2026_10_22/H4_v3_saddle_stability/README.md @ 58bf38dcb3; covariant_push_2026_10_22/C5_health/c5_health_results.json @ f3e7d4d661

Lanes: C5 / H4 / X3

Leaf problem well-posed

CONDITIONAL

On the board: C8's (P, lapse) leaf problem well-posed (H4 §5); v3s: switched leaf functional convex, unique; obstacle-type free boundary (X3); open: infinite volume, coupled GR + dust



source: covariant_push_2026_10_22/C8_instantaneous/c8_instantaneous_results.json (K1, K4, K5); v3s switch: rebuild_2026_10_22/X3_switched_stiffness/x3_switched_stiffness_results.json (B3S)

ON THE PLATE

C8's numerical uniqueness checks of the (P, lapse) leaf problem: spherical point-mass solutions at r_M/L* = 0.3 and 3 from two initial guesses and on a doubled domain (K4), and a 26^3 3D two-mass solve from two guesses (K5).

HOW IT IS PLATED

Horizontal log bars of the relative difference between independently obtained solutions, each with its frozen pass line as a red tick; all bars sit well left of their ticks.

WHY IT IS CONDITIONAL

On each khronon leaf the P and lapse equations are the Euler-Lagrange equations of one strictly convex functional (K2), with principal symbol bounded below by min(K, gamma_L xi^2) (K1, min ratio 1), so on a bounded domain the solution exists and is unique. The numbers confirm it: two guesses agree to 8.7e-14 and 3.9e-15 in spherical cases and to 3.7e-8 / 1.4e-7 in 3D. v3s's switched functional stays convex with an obstacle-type free boundary (X3 B3-S). It stays conditional because the infinite-volume theorem and the coupled GR + dust system are open.

Sources: covariant_push_2026_10_22/C8_instantaneous/c8_instantaneous_results.json @ aa113a5e96; covariant_push_2026_10_22/C8_instantaneous/README.md @ aa113a5e96; rebuild_2026_10_22/X3_switched_stiffness/x3_switched_stiffness_results.json @ 31febd3b70; rebuild_2026_10_22/H4_v3_saddle_stability/README.md @ 58bf38dcb3

Lanes: C8 / H4 / X3

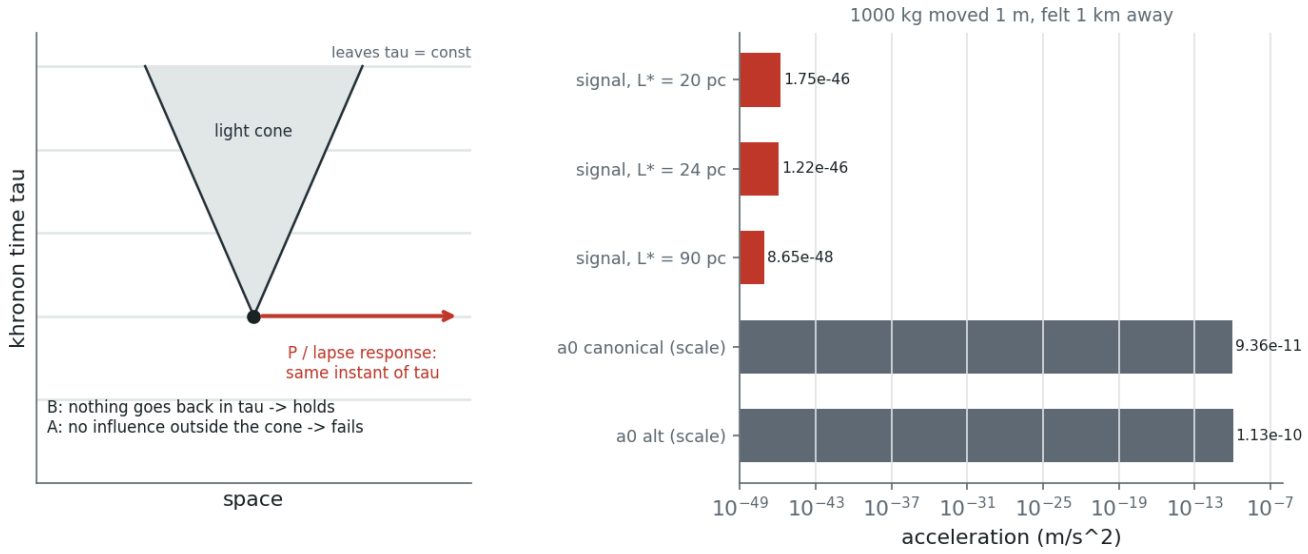
Causality: criterion A / B

CONDITIONAL

On the board: instantaneous constraints; faster-than-light signal ~1e-46 m/s^2

Causality: criterion A / B

CONDITIONAL: ordered in khronon time (B holds); faster than light in principle (A fails), at ~1e-46 m/s^2



source: covariant_push_2026_10_22/C8_instantaneous/c8_instantaneous_results.json (signal_illustration); a0 from covariant_push_2026_10_22/C6_literal_cassini/c6_literal_cassini_results.json (inputs)

ON THE PLATE

C8's illustrative lab signal: the change in P force 1 km away when 1000 kg moves 1 m, at L* = 20, 24, 90 pc; the a0 footings from C6's inputs for scale.

HOW IT IS PLATED

Left: schematic of khronon leaves (horizontal lines), a light cone, and the P/lapse response running along the leaf at the same instant of tau. Right: log bars of the signal acceleration against a0 on both footings.

WHY IT IS CONDITIONAL

The P and lapse fields are fixed by an elliptic problem on each leaf, so a moving source changes the P force elsewhere in the same instant of khronon time. Every influence goes forward in tau or stays in a leaf, so criterion B (no paradox, one global time) holds; criterion A (nothing outside the light cone) fails in principle, as a real signal and not a gauge artefact. The size is the qualifier: 1.2e-46 m/s^2 at L* = 24 pc, about 36 orders below a0. No test identified can see it.

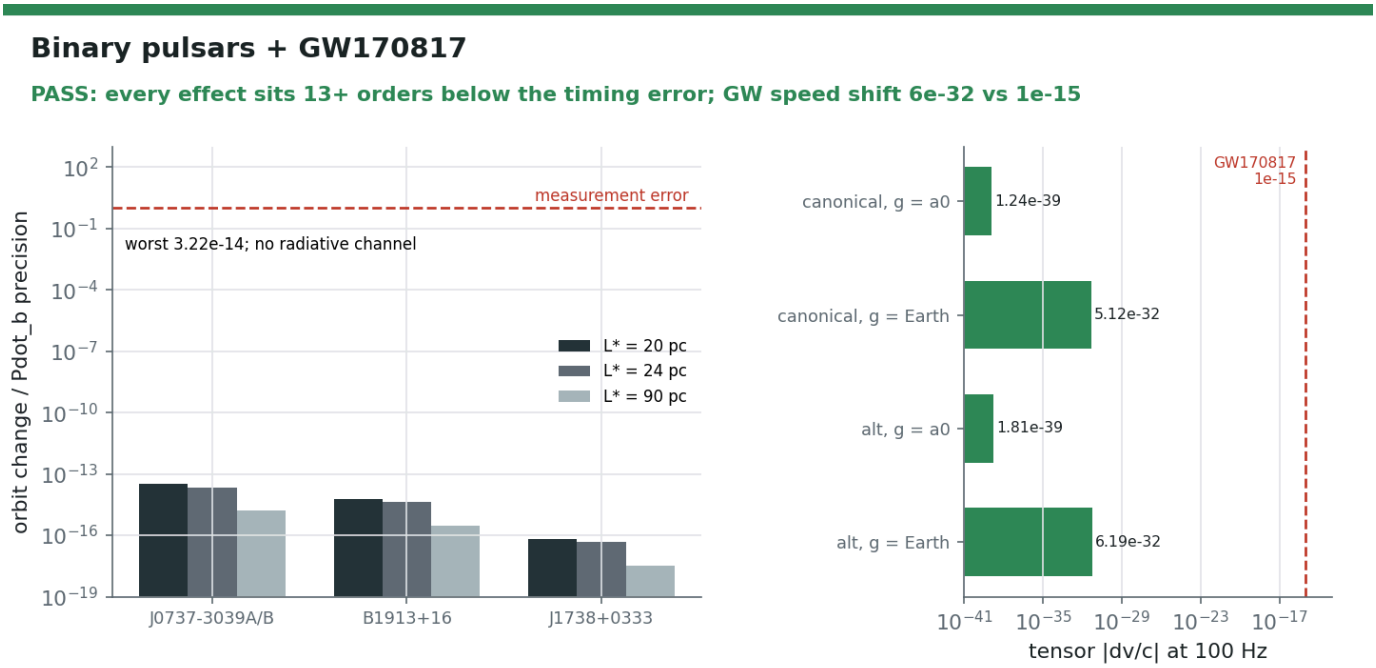
Sources: covariant_push_2026_10_22/C8_instantaneous/c8_instantaneous_results.json @ aa113a5e96; covariant_push_2026_10_22/C8_instantaneous/README.md @ aa113a5e96; covariant_push_2026_10_22/C6_literal_cassini/c6_literal_cassini_results.json @ ae428dc252

Lanes: C5 / C8

Binary pulsars + GW170817

PASSES

On the board: no radiative channel; <= 3e-14 of double-pulsar error



source: covariant_push_2026_10_22/C8_instantaneous/c8_instantaneous_results.json (K7, K9); Pdot_b precisions are record transcriptions

ON THE PLATE C8 K7: the fractional orbit change eps = (a/L*)^2 for J0737-3039A/B, B1913+16 and J1738+0333 at L* = 20/24/90 pc, divided by each system's Pdot_b precision (record transcriptions of Kramer+2021, Weisberg & Huang 2016, Freire+2012, not re-downloaded). C8 K9: tensor speed shift at 100 Hz.

HOW IT IS PLATED Left: grouped log bars of orbit change over Pdot_b precision for the three pulsars and three L*, with 1 (the measurement error) dashed. Right: log bars of tensor |dv/c| on both footings and two background fields against the GW170817 bound 1e-15.

WHY IT PASSES At beta = eta = 0 there is no propagating non-tensor mode at these scales, so there is no dipole or extra radiation; the instantaneous force does zero work per orbit. The only effect is a conservative orbit change, at most 3.22e-14 of the double pulsar's timing error. The tensor speed shift from the P background is at most 6.19e-32 against 1e-15. P3 adds that the pair-metric mixing at the double pulsar is 2.0e-9, so the leakage stays negligible.

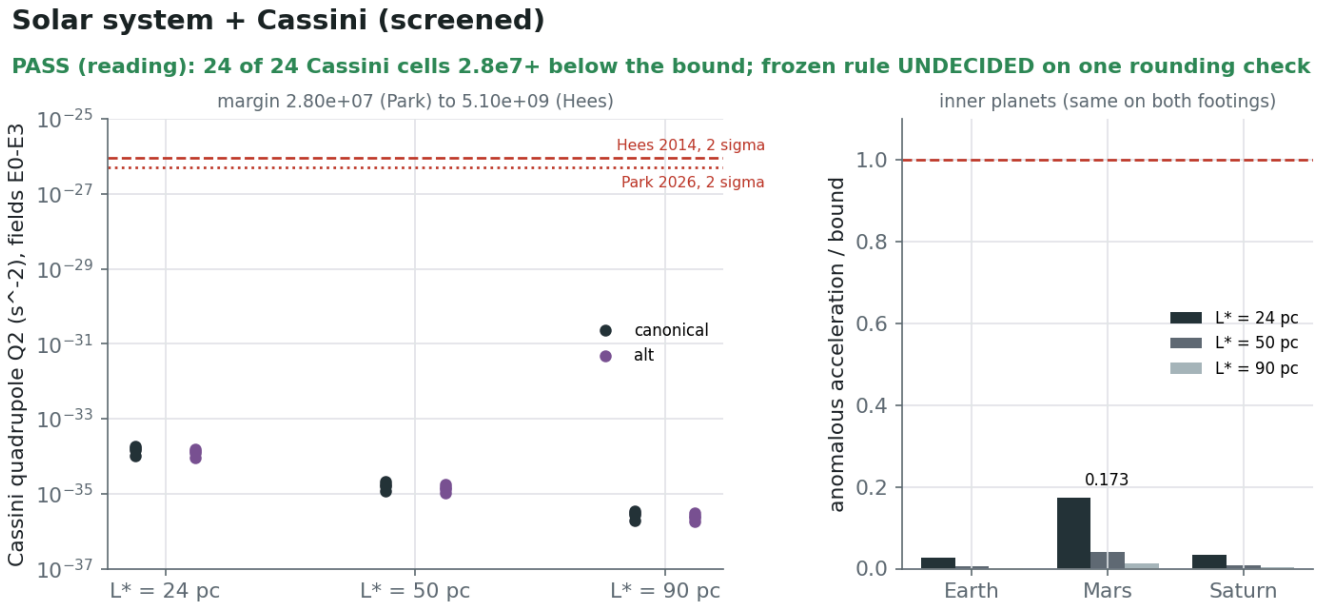
Sources: covariant_push_2026_10_22/C8_instantaneous/c8_instantaneous_results.json @ aa113a5e96; covariant_push_2026_10_22/C8_instantaneous/README.md @ aa113a5e96; rebuild_2026_10_22/P3_v3_strong_field/p3_v3_strong_field_results.json @ 2b067f2813

Lanes: C8

Solar system + Cassini (screened)

PASSES

On the board: Q2 2.8e7-5.1e9 inside bound at L 24-90 pc; Mars 0.17; C6's frozen rule UNDECIDED on one rounding check*



source: covariant_push_2026_10_22/C6_literal_cassini/c6_literal_cassini_results.json (R1, R2, checks C2)

ON THE PLATE C6: the Cassini quadrupole Q2 of the healthy C4-literal screened theory (gamma_F = 0, AQUAL) in a uniform Galactic field, 24 cells = 2 footings x L* 24/50/90 pc x fields E0-E3; inner-planet anomalous accelerations against their bounds.

HOW IT IS PLATED Left: log scatter of Q2 per cell, grouped by L* and coloured by footing, with the Hees 2014 (9e-27) and Park 2026 (5.2e-27) 2-sigma upper lines. Right: bars of anomalous acceleration over bound for Earth, Mars, Saturn at the three L*.

WHY IT PASSES Inside L* the P field is a hedgehog with no quadrupole at leading order, so Q2 is 1e-34 to 1e-36, between 2.80e7 (Park, worst) and 5.10e9 (Hees, best) below the bounds. The planets see only the hedgehog GM/L*^2; the tightest is Mars at 0.173 of its bound (L* = 24 pc). By the lane's frozen rule the verdict is UNDECIDED because check C2 failed on float rounding (3.8e-11 against a 1e-12 line), not physics; the board reads it as PASS and says so.

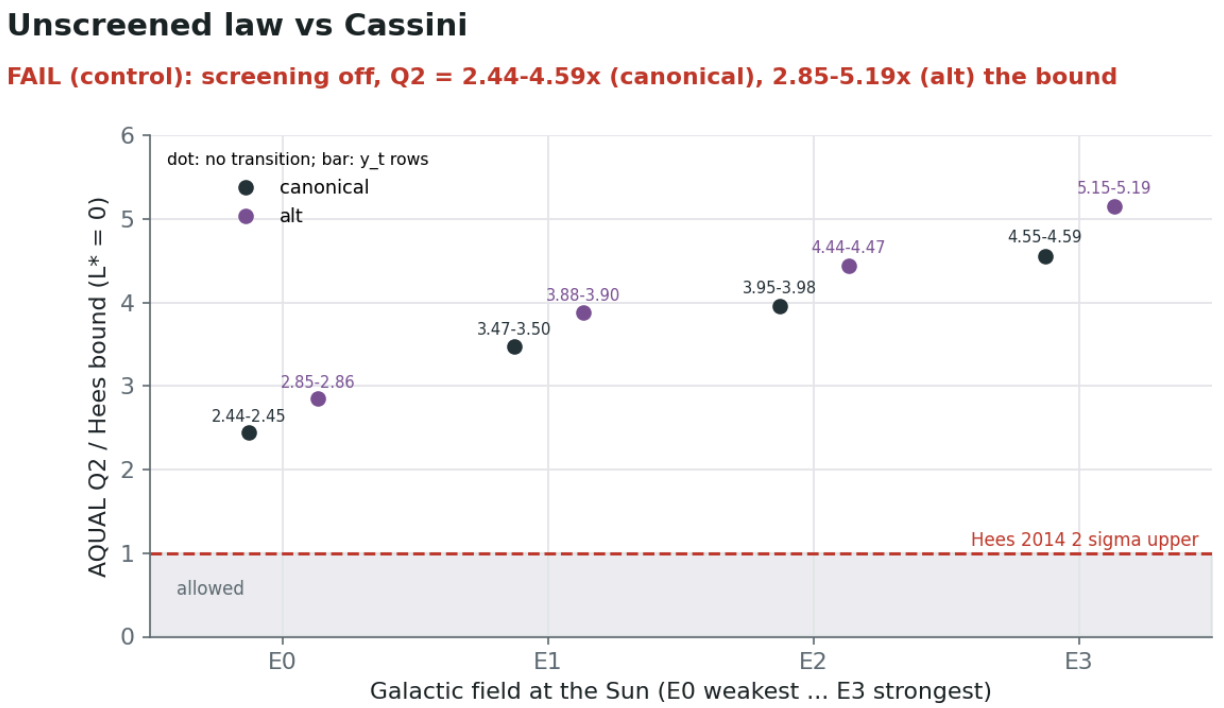
Sources: covariant_push_2026_10_22/C6_literal_cassini/c6_literal_cassini_results.json @ ae428dc252

Lanes: C6

Unscreened law vs Cassini

FAILS

On the board: control: screening switched off; fails 2.4-5.2x, so L > 0 is required*



source: doors_push_2026_10_22/E2_field_equation/e2_field_equation_results.json (R1, R1_outcome)

ON THE PLATE E2: Cassini Q2 of v3's own AQUAL equation (mu = x/(1+x)) with screening switched off (L* = 0), both footings, fields E0-E3, five transition settings y_t (none, 60, 66.38, 77.88, 140), over the Hees 2014 2-sigma upper 9e-27.

HOW IT IS PLATED Points: Q2 / bound per field and footing (dot = no-transition row; faint bar = spread over y_t rows, too small to see), labelled with the min-max over rows; the allowed region below 1 is shaded.

WHY IT FAILS This is a control, not the theory's claim. Without screening, the Galactic external field leaves a quadrupole at Saturn's orbit 2.44-4.59 times (canonical) and 2.85-5.19 times (alt) the bound in every cell, so the plain law fails Cassini. That failure is why L* > 0 is required; the screened version passes (previous tile).

Sources: doors_push_2026_10_22/E2_field_equation/e2_field_equation_results.json @ 474050bcd8

Lanes: E2

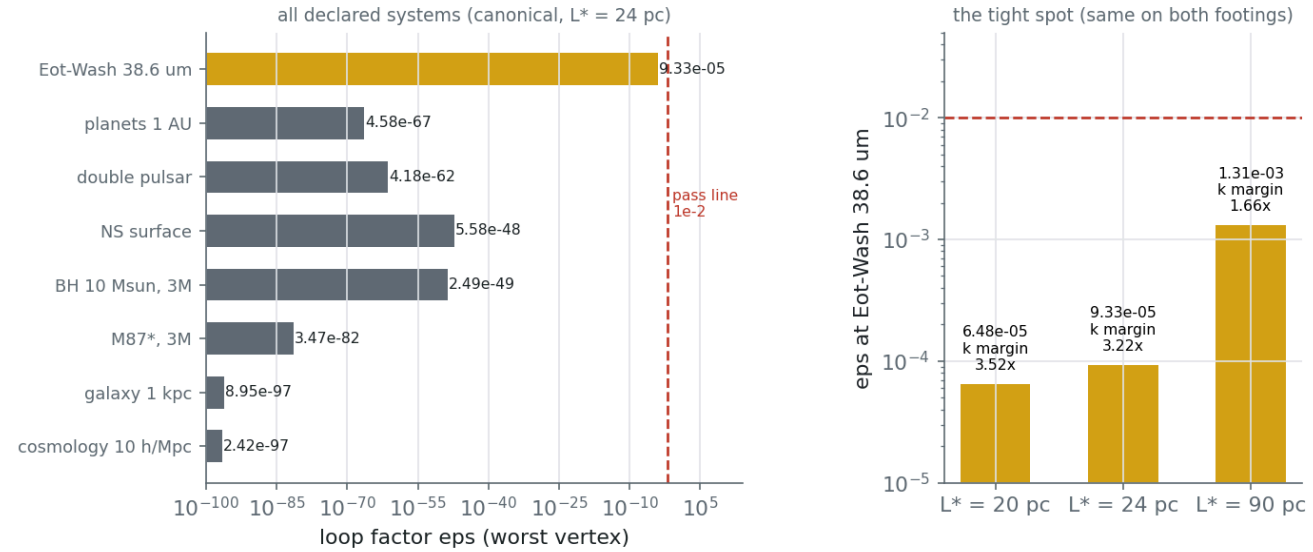
Strong coupling

CONDITIONAL

On the board: estimate: loop factor ≤ 1.3e-3; margin only 1.7× in k at 38.6 μm; sub-5 μm tests would probe it

Strong coupling

CONDITIONAL (estimate): weakly coupled at every declared system; only a 1.7x margin in k at Eot-Wash



source: rebuild_2026_10_22/P3_v3_strong_field/p3_v3_strong_field_results.json (E4); pass line eps <= 1e-2 from rebuild_2026_10_22/P3_v3_strong_field/FROZEN_CRITERIA.md V4

ON THE PLATE

P3 E4: the decoupling-limit loop factor eps = (c_pi k / Lambda)^4 of the worst vertex on each declared background (lab Eot-Wash scale 38.6 um, planets, double pulsar, NS, BH, M87*, galaxy, cosmology) at L* = 20/24/90 pc, both footings.

HOW IT IS PLATED

Left: log bars of eps per system (canonical, 24 pc) against the frozen pass line 1e-2. Right: eps at Eot-Wash for the three L* with the margin in k, (1e-2/eps)^(1/4), printed on each bar.

WHY IT IS CONDITIONAL

Everywhere except the lab the pair is extremely weakly coupled (eps below 1e-47). The Eot-Wash 38.6 um scale is the tight spot: eps = 1.31e-3 at L* = 90 pc, a margin of only 1.66 in k (7.6 in eps), the same on both footings. Gravity tests below about 5 um would probe the pair's strong coupling. It is an estimate (decoupling limit, first order in background gradients, O(1) loop factors), so the tile stays conditional.

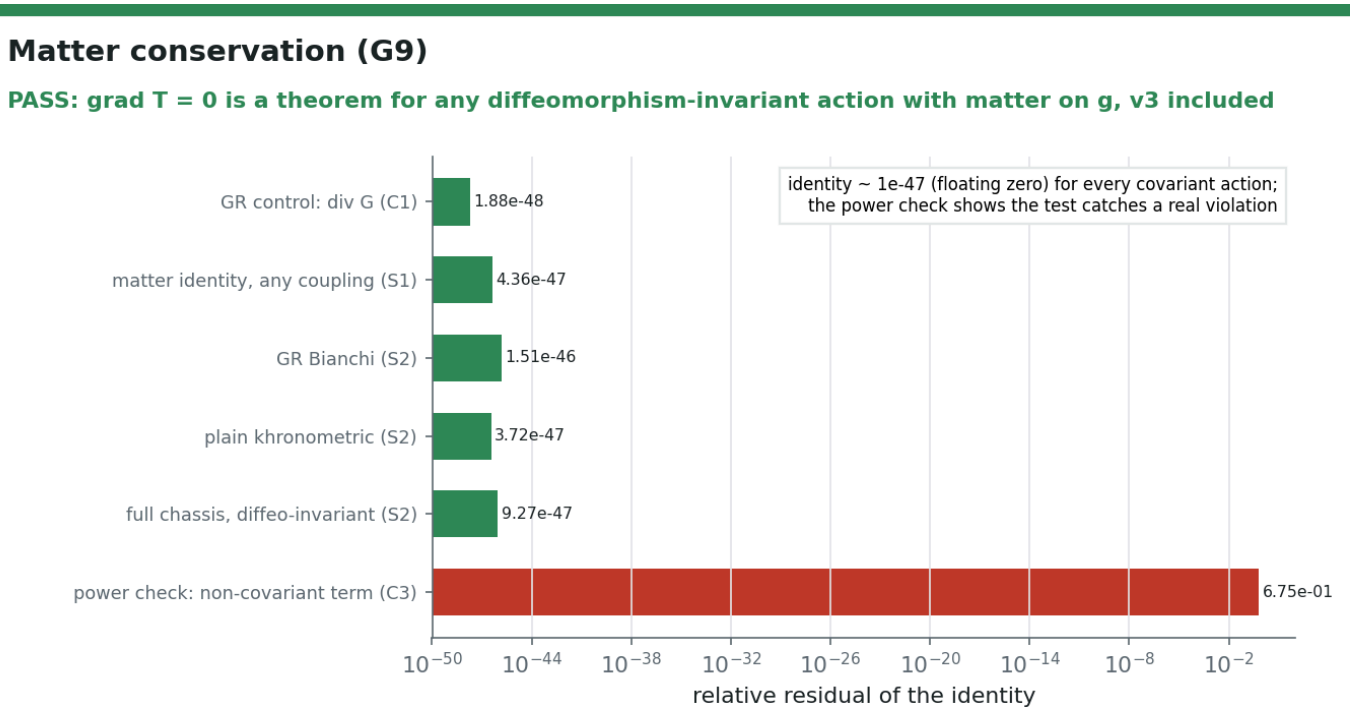
Sources: rebuild_2026_10_22/P3_v3_strong_field/p3_v3_strong_field_results.json @ 2b067f2813; rebuild_2026_10_22/P3_v3_strong_field/FROZEN_CRITERIA.md @ 988eb4940d

Lanes: P3

Matter conservation (G9)

PASSES

On the board: Bianchi identity, grad T = 0: holds for any diffeomorphism-invariant action, v3 included



source: campaign_fresh_gravity/CFG329_g9_g0_structure/cfg329_g9_g0_results.json (numbers C1, S1, S2); v3 is diffeo-invariant per covariant_push_2026_10_22/C4_khronon_parent

ON THE PLATE CFG329's symbolic identity checks: relative residuals of the generalised Bianchi identity and the matter identity grad T = 0 at random points, for GR, the plain khronometric theory and the full (old) chassis, plus a deliberately non-covariant power check.

HOW IT IS PLATED Horizontal log bars of relative residual: every covariant case sits at 1e-46 to 1e-48 (floating zero); the non-covariant control (red) is 0.675, showing the test can see a violation.

WHY IT PASSES grad T = 0 is a Noether-II identity: it holds for any diffeomorphism-invariant action whose matter couples to the metric alone. CFG329 computed it on the older chassis, but the result does not depend on the chassis, and v3 (C4's Khronon parent) is diffeomorphism-invariant with minimally coupled matter, so it carries over. The power check (a coordinate-time term) gives an O(1) residual, so the near-zero values are a real pass, not a blind test.

Sources: campaign_fresh_gravity/CFG329_g9_g0_structure/cfg329_g9_g0_results.json @ 86085d6efa; covariant_push_2026_10_22/C4_khronon_parent/c4_khronon_parent_results.json @ 5f4f845554

Lanes: CFG329 / C4

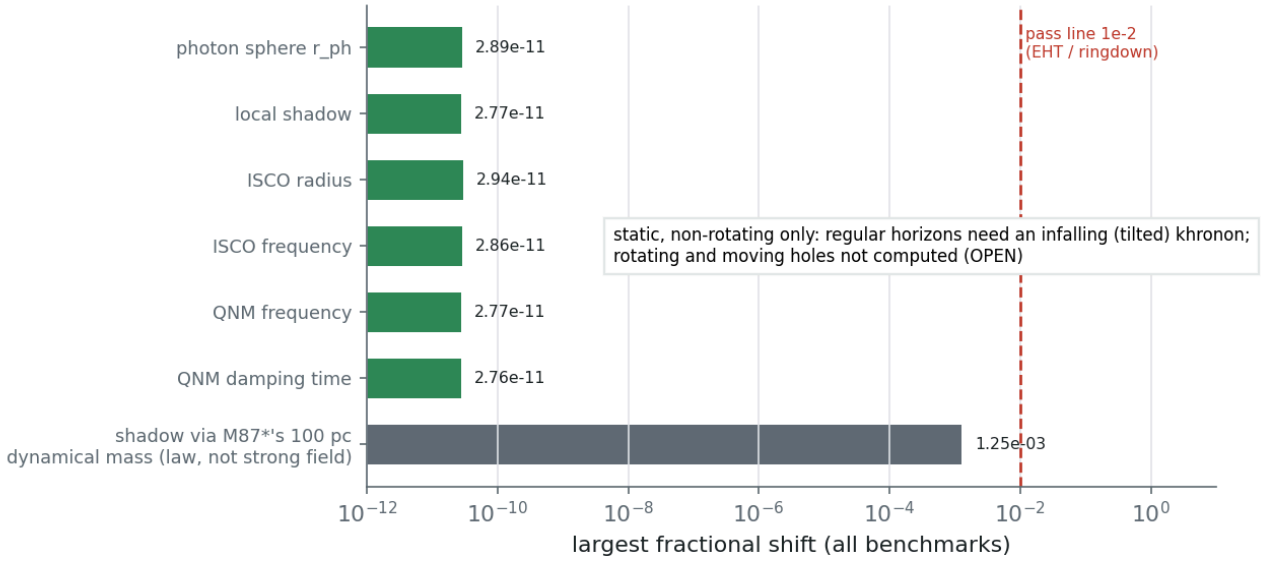
Black holes (EHT, LIGO ringdown)

CONDITIONAL

On the board: PASS for static non-rotating holes (shifts ≤ 3e-11); regular holes need an infalling khronon; rotating/moving holes open

Black holes (EHT, LIGO ringdown)

CONDITIONAL: static holes pass (shape shifts <= 3e-11); rotating and moving holes open



source: rebuild_2026_10_22/P3_v3_strong_field/p3_v3_strong_field_results.json (worst_shifts)

ON THE PLATE

P3: regular stationary, non-rotating v3 black-hole exteriors (M87*, Sgr A*, 10 and 63 Msun; both footings; L* 20/24/90 pc; charge c grid), and the largest fractional shift of each strong-field observable over all benchmarks.

HOW IT IS PLATED

Horizontal log bars of the worst shift for photon sphere, shadow, ISCO radius and frequency, QNM frequency and damping (green), and the law's effect through M87*'s 100 pc dynamical mass (grey), against the 1e-2 pass line.

WHY IT IS CONDITIONAL

Regular horizons exist only with an infalling (tilted) khronon; on the static foliation the dust stress diverges at the horizon. On the regular members every strong-field shape shift is at most 2.94e-11 and the QNM shifts at most 2.77e-11, far below 1e-2. The largest number, 1.25e-3, is not strong-field: it is the law acting on M87*'s stellar-dynamics mass at 100 pc (illustrative). The tile is conditional because rotating and moving holes are not computed, and the SMBH khronon charge is not fixed.

Sources: rebuild_2026_10_22/P3_v3_strong_field/p3_v3_strong_field_results.json @ 2b067f2813; rebuild_2026_10_22/P3_v3_strong_field/FROZEN_CRITERIA.md @ 988eb4940d

Lanes: P3

~ Mains ~

Galaxies · 10 dishes

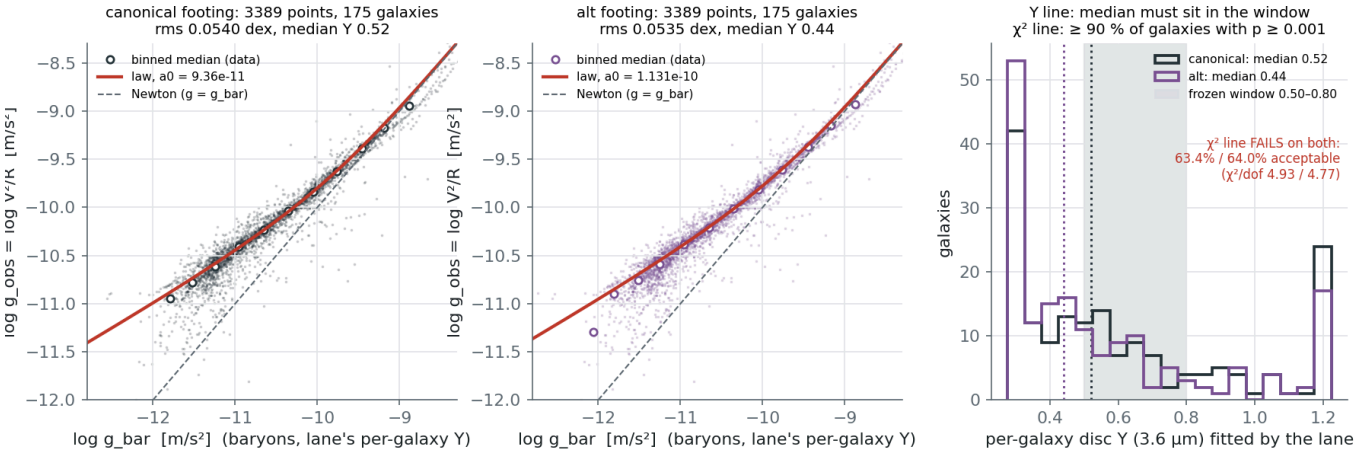
Rotation curves (SPARC RAR)

CONDITIONAL

On the board: one Y per galaxy, EFE solved (109): rms 0.054/0.054 dex; median Y 0.52 / 0.44 (alt below the 0.50–0.80 window); χ^2 line fails both (63–64% acceptable; ν_mono same)

Rotation curves: the SPARC radial-acceleration relation

CONDITIONAL: canonical passes the rms + Y line, alt's median Y 0.44 falls below 0.50; the χ^2 line fails on both



source: rebuild_2026_10_22/P4_v3_galaxy_tiles/p4_v3_galaxy_tiles_results.json (per-galaxy Y, L★ 24 pc); real_research/data/sparc_data

ON THE PLATE

175 SPARC galaxies (Lelli+16 rotmod files): observed acceleration V^2/R against the baryonic acceleration from the gas, disc and bulge velocities. P4 fitted one disc Y per galaxy (bulge 1.4Y) and solved v3's AQUAL law with each galaxy's external field (Chae+21 environments, 109 of 175) and screening at L★ = 24 and 90 pc.

HOW IT IS PLATED

Left and middle: every measured point (small dots) and binned medians (open circles) against v3's law on the canonical and the alt footing (red line), with Newton as the dashed diagonal. Right: the distribution of the fitted per-galaxy Y on both footings, with the frozen 0.50–0.80 window shaded; P4's Y grid runs 0.30–1.20, so the piles at the two ends are galaxies sitting on the grid bounds.

WHY IT IS CONDITIONAL

The points follow the law closely on both footings: the rms scatter is 0.054 dex, well inside the 0.110 line. On the alt footing the larger a_0 makes the fits prefer lighter discs, so the median Y is 0.44, below the frozen 0.50 window; that splits the tile, so it is conditional. The χ^2 line fails on both footings (about 63–64 % of galaxies acceptable against 90 %), and the record's kernel ν_mono does the same with the same freedom, so the miss is the error budget or non-circular motions, not something specific to v3. Points are drawn with the isolated law at the file's per-galaxy Y; the external field changes the rms by under 0.001 dex.

Sources: rebuild_2026_10_22/P4_v3_galaxy_tiles/p4_v3_galaxy_tiles_results.json @ 6c0249a8bc; rebuild_2026_10_22/P4_v3_galaxy_tiles/README.md @ 6c0249a8bc; real_research/data/sparc_data @ 999f94dfee

Lanes: P4

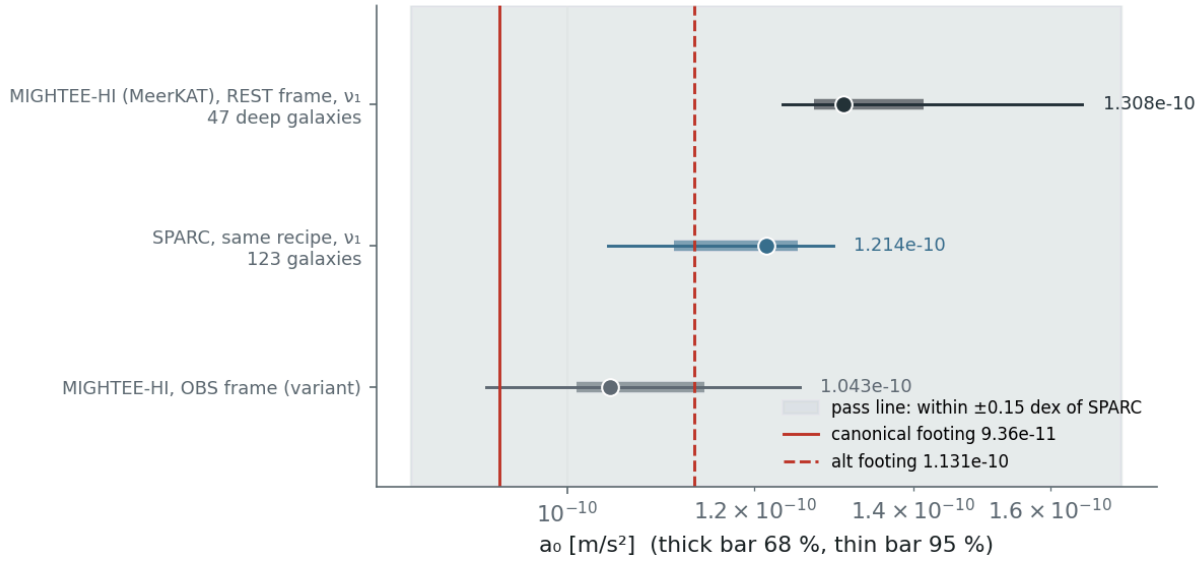
Local a0 from MeerKAT

PASSES

On the board: a0 1.31e-10 vs SPARC same recipe 1.21e-10 (+0.032 dex, line 0.15)

Local a_0 from MeerKAT (MIGHTEE-HI)

PASS: MeerKAT sits +0.032 dex from SPARC's same-recipe value, against a ±0.15 dex line



source: rebuild_2026_10_22/P4_v3_galaxy_tiles/p4_v3_galaxy_tiles_results.json tiles.T2.rows

ON THE PLATE — 47 MIGHTEE-HI (MeerKAT) galaxies, all deep in the low-acceleration regime, with HI line widths in the rest frame (the frame CFG309 decided); a_0 estimated with the same recipe as 123 SPARC galaxies, using v3's kernel ν_1.

HOW IT IS PLATED — A log a_0 axis. Each row is an estimate with its 68 % (thick) and 95 % (thin) bootstrap interval: MeerKAT rest frame, SPARC same recipe, and the observed-frame variant in grey. The grey band is the ±0.15 dex pass line around SPARC; red lines mark the two footings.

WHY IT PASSES — The MeerKAT value lands 0.03 dex above SPARC's same-recipe value, far inside the 0.15 dex line, so the tile passes. The observed-frame variant (−0.066 dex) also passes, so the verdict does not hinge on the frame. Both footings sit outside MeerKAT's statistical 95 % interval but inside the recipe's width; the flux-scale systematic is not addressed. No environment estimate exists for these galaxies, so they are treated as isolated.

Sources: rebuild_2026_10_22/P4_v3_galaxy_tiles/p4_v3_galaxy_tiles_results.json @ 6c0249a8bc;
rebuild_2026_10_22/P4_v3_galaxy_tiles/README.md @ 6c0249a8bc;
campaign_fresh_gravity/CFG309_mightee_width_frame/README.md @ 53f8fa937e

Lanes: CFG301 / CFG309 / P4

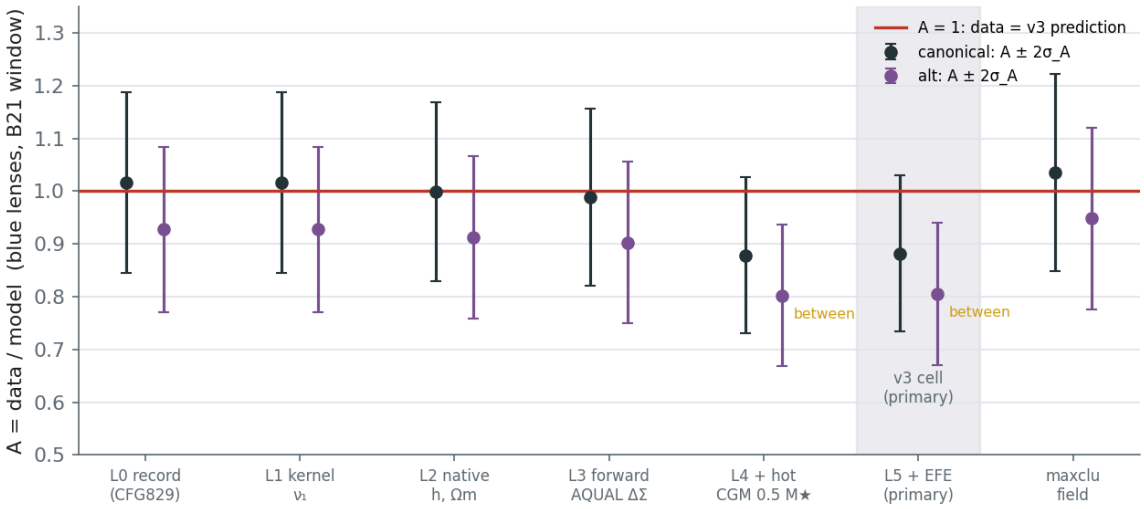
Isolated-galaxy lensing, inner

CONDITIONAL

On the board: v3 blue A 0.88 / 0.81 (primary field); EFE ≤ 0.004 here; decided by hot CGM and M★; holds for μ_K < 0.1/Mpc (S4)

Isolated-galaxy lensing, inner window (blue lenses)

CONDITIONAL: A 0.88 (canonical, consistent) / 0.81 (alt, between); the hot CGM step is what moves it



source: rebuild_2026_10_22/S2_v3_lowg_lensing/s2_v3_lowg_lensing_results.json ladder, cells_central

ON THE PLATE

KiDS galaxy–galaxy lensing of isolated blue (late-type) lenses in Brouwer+21's low-acceleration window (g_bar 10^−13.3–10^−11.9 m/s², 6 points), forward-modelled as v3's ΔΣ over the record's 181,477-lens stack.

HOW IT IS PLATED

The amplitude A = data/model (with 2σ_A bars) at each step of the lane's ladder, from the record's algebraic number to the v3 cell (AQUAL, hot CGM, external field), and at the maximum-cluster field. The red line A = 1 means data equal the prediction; 'consistent' is |A − 1| ≤ 2σ_A.

WHY IT IS CONDITIONAL

Each ingredient of v3 is added one at a time. The kernel, cosmology and forward-modelling barely move A; adding a hot circumgalactic gas halo of half the stellar mass drops it by about 0.11, and the external field moves it by at most 0.004 in this window. On the canonical footing A stays consistent with 1; on the alt footing it is 'between' (−2.9σ) and only becomes consistent at the maximum-cluster field, so the tile is conditional. It hinges on the unmeasured hot gas of blue lenses and on SPS stellar masses, and it holds only for μ_K < 0.1/Mpc (S4); the $\mu_K^2\psi$ term was not included in S2.

Sources: rebuild_2026_10_22/S2_v3_lowg_lensing/s2_v3_lowg_lensing_results.json @ 0a3db18162;
rebuild_2026_10_22/S2_v3_lowg_lensing/README.md @ 0a3db18162;
rebuild_2026_10_22/S4_v3_muK_clusters/s4_v3_muk_clusters_results.json @ 0ef14fdd9a

Lanes: S2 / S4

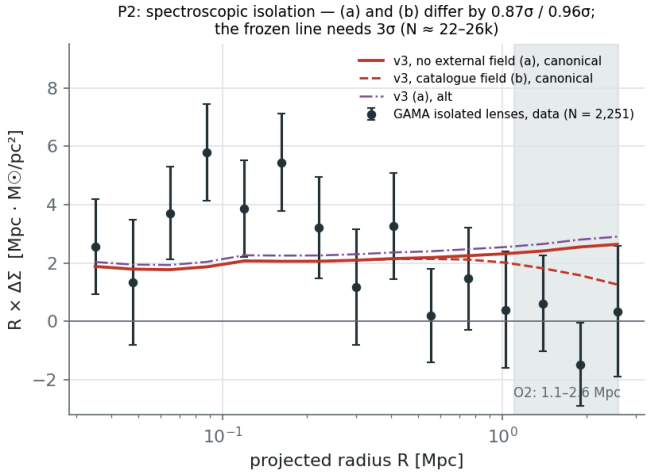
Isolated-galaxy lensing, all radii

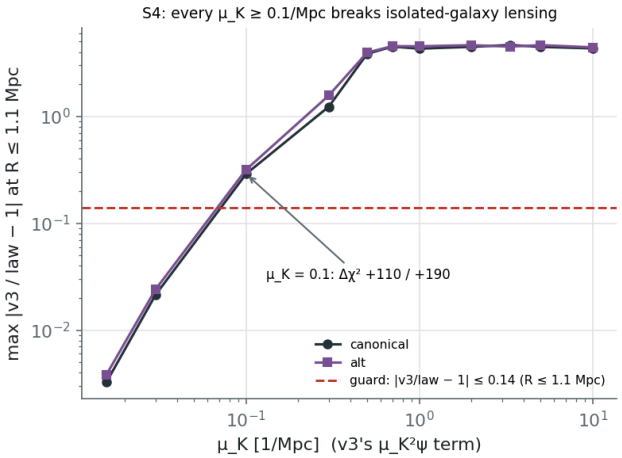
NOT DECIDABLE YET

On the board: spectroscopic isolation (P2): N too small (needs ~22–26k lenses); data favour the catalogue's EFE; S4: the μ_K² term ruins it for every μ_K ≥ 0.1/Mpc (Δχ² +110 / +190 at 0.1)

Isolated-galaxy lensing at all radii

UNDECIDED: the clean sample is too small to tell v3's external-field readings apart; the μ_K² term must stay below 0.1/Mpc





source: rebuild_2026_10_22/P2_spec_isolated_lensing/p2_spec_isolated_lensing_results.json verdict.profile; rebuild_2026_10_22/S4_v3_muK_clusters/s4_v3_muk_clusters_results.json galaxy

ON THE PLATE

Weak lensing out to 3 Mpc around 2,251 spectroscopically isolated GAMA galaxies (P2), plus S4's test of v3's μ_K²ψ term against Brouwer+21's isolated-lens profile with its released covariance.

HOW IT IS PLATED

Left: the measured excess surface density ΔΣ (multiplied by R so the outer points show) with errors against v3's prediction with no external field (solid red) and with the field the catalogue's own baryons imply (dashed), canonical footing; the alt no-field curve is dash-dot. Right: the largest departure of v3 from the plain law inside 1.1 Mpc as μ_K grows, against the 0.14 guard line.

WHY IT IS NOT DECIDABLE YET

Beyond 1.1 Mpc the stack sits below the zero-field prediction and is consistent with the catalogue-field one, but the two predictions differ by under 1σ with this sample; the frozen line needs 3σ, which needs roughly ten times more lenses, so the tile is undecided. Separately, S4 finds that any μ_K of 0.1/Mpc or more makes v3 depart from the law by more than the guard inside 1.1 Mpc (Δχ² +110 / +190 at 0.1), so the tile can only stay open if μ_K is below that.

Sources: rebuild_2026_10_22/P2_spec_isolated_lensing/p2_spec_isolated_lensing_results.json @ 1ebfa54245; rebuild_2026_10_22/P2_spec_isolated_lensing/README.md @ 1ebfa54245; rebuild_2026_10_22/S4_v3_muK_clusters/s4_v3_muk_clusters_results.json @ 0ef14fdd9a

Lanes: CFG902 / CFG903 / F2 / P2 / S4

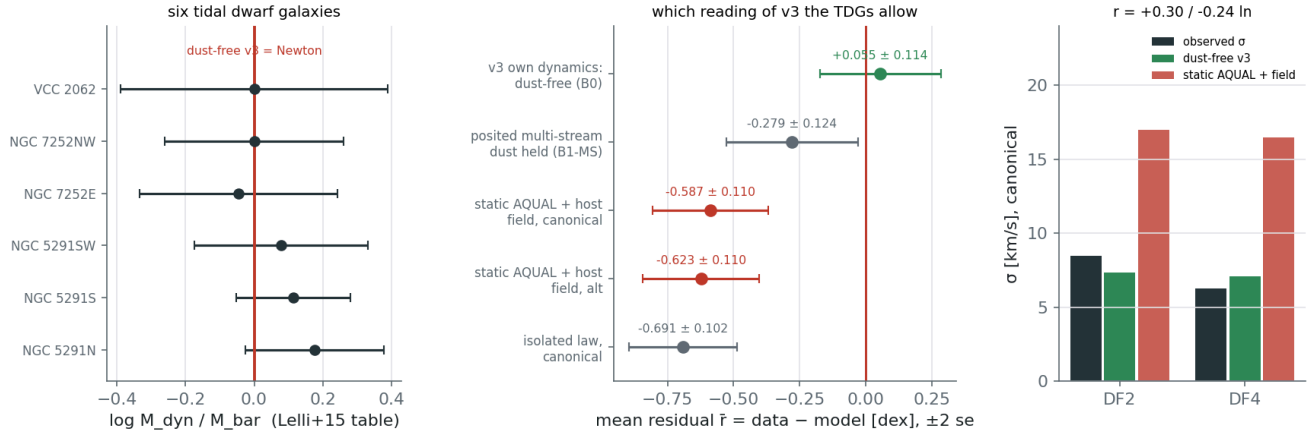
Tidal dwarfs, NGC 1052-DF2/DF4

PASSES

On the board: static AQUAL (P4) read −0.59 dex; v3's own dynamics (T3) makes hosted systems dust-free → Newtonian: TDGs +0.055 ± 0.114 dex, DF2/DF4 +0.30/−0.24 ln; no knob

Tidal dwarfs and NGC 1052-DF2 / DF4

PASS (native, no knob): hosted systems that fall with their host carry no held dust, so v3 predicts Newton, and the data sit there



source: rebuild_2026_10_22/U1_v3_dust_content/u1_v3_dust_content_results.json numbers.N5.tdg; rebuild_2026_10_22/P4_v3_galaxy_tiles/p4_v3_galaxy_tiles_results.json T5; real_research/data/tidal_dwarfs/lelli2015_tdg.csv

ON THE PLATE Six tidal dwarf galaxies (Lelli+15: rotation, baryonic and dynamical masses) and the two ultra-diffuse galaxies NGC 1052-DF2 and DF4 (velocity dispersions), all sitting in their hosts' fields (g_e 0.15–0.28 a_0).

HOW IT IS PLATED Left: each TDG's measured dynamical-to-baryonic mass ratio (log) with its error; the red line at zero is what dust-free v3 (Newton) predicts. Middle: the mean residual with ±2 se for each reading of v3 — dust-free (U1's own dynamics), posited multi-stream dust, the static AQUAL solution with the host field (P4), and the isolated law. Right: DF2 / DF4 observed σ against dust-free v3 and static AQUAL.

WHY IT PASSES Solved as a static AQUAL problem in the host's field, v3 over-predicts these systems: the TDGs sit −0.59 dex below it and DF2/DF4 −1.66 ln below (P4, a fail). U1 then showed from v3's own equations that a system falling freely with its host cannot hold dust (the host's pull on the dust is exactly the external-field term), so v3's own prediction is Newtonian. The data sit on that: TDGs +0.055 ± 0.114 dex, DF2/DF4 +0.30 / −0.24 ln, with no knob. TDG discs have turned less than one orbit, and every estimator assumes equilibrium.

Sources: rebuild_2026_10_22/U1_v3_dust_content/u1_v3_dust_content_results.json @ bbb6bfcc71;
rebuild_2026_10_22/U1_v3_dust_content/README.md @ 58de634d74;
rebuild_2026_10_22/P4_v3_galaxy_tiles/p4_v3_galaxy_tiles_results.json @ 6c0249a8bc;
rebuild_2026_10_22/P4_v3_galaxy_tiles/p4_v3_galaxy_tiles.out @ 6c0249a8bc;
rebuild_2026_10_22/P4_v3_galaxy_tiles/README.md @ 6c0249a8bc;
real_research/data/tidal_dwarfs/lelli2015_tdg.csv @ 3bdfe877d9

Lanes: P4 / U1

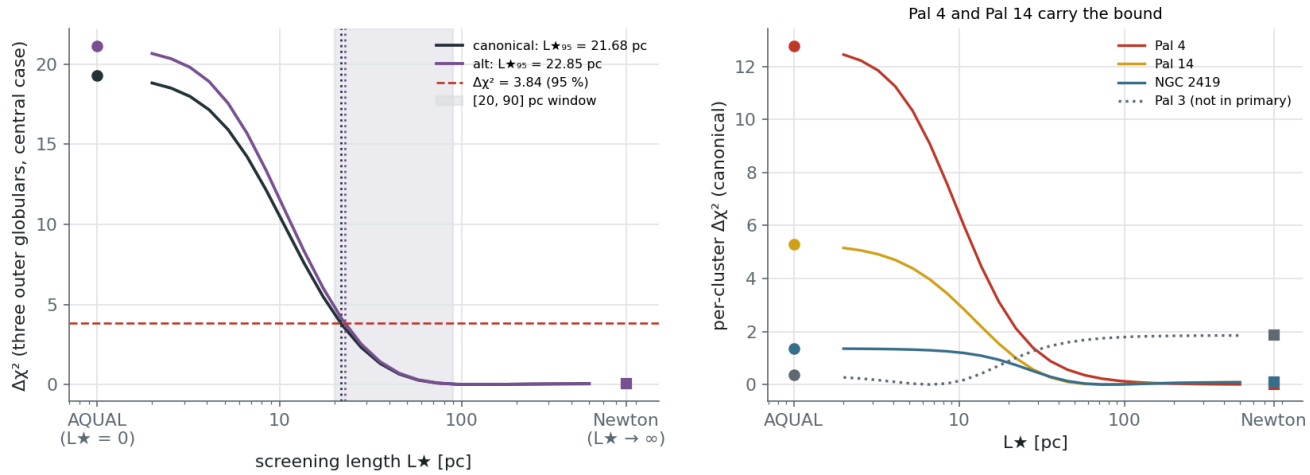
Outer globulars: L* floor

CONDITIONAL

On the board: L >= 21.7 / 22.9 pc (SPS M/L); unscreened law excluded*

Outer globular clusters: the L★ screening floor

CONDITIONAL: unscreened AQUAL excluded; floor L★ ≥ 21.68 / 22.85 pc with SPS M/L, but the M/L band reaches 2.93 pc



source: doors_push_2026_10_22/D10_native_Lstar/d10_native_lstar_results.json chi2_curve, verdict

ON THE PLATE

Line-of-sight velocity dispersions of the distant Milky Way globulars NGC 2419, Pal 4 and Pal 14 (Pal 3 reported), each solved as a Plummer sphere in the Galaxy's field with v3's stiffened field equation; SPS stellar M/L (Y_V logN(1.6, 0.15 dex)), binary fraction 0.3.

HOW IT IS PLATED

Left: $\Delta\chi^2$ against the screening length L★ on both footings, from unscreened AQUAL (left end) to Newton (right end); the red dashed line is the 95 % cut and the dotted verticals the resulting floors. Right: each cluster's own $\Delta\chi^2$ (canonical), showing which clusters set the bound.

WHY IT IS CONDITIONAL

Unscreened AQUAL predicts dispersions well above the measured ones for Pal 4 and Pal 14, so it is excluded in every cell; screening with L★ of about 22 pc or more brings v3 back to the data, and no finite L★ beats Newton, so this is a lower bound only. The floor is set by the M/L prior: with the mass-function-slope M/L (CFG332) the band drops to 3–10 pc and straddles the 20 pc window edge, which is why the tile is conditional.

Sources: doors_push_2026_10_22/D10_native_Lstar/d10_native_lstar_results.json @ 3ac0573063; doors_push_2026_10_22/D10_native_Lstar/README.md @ 3ac0573063

Lanes: D10

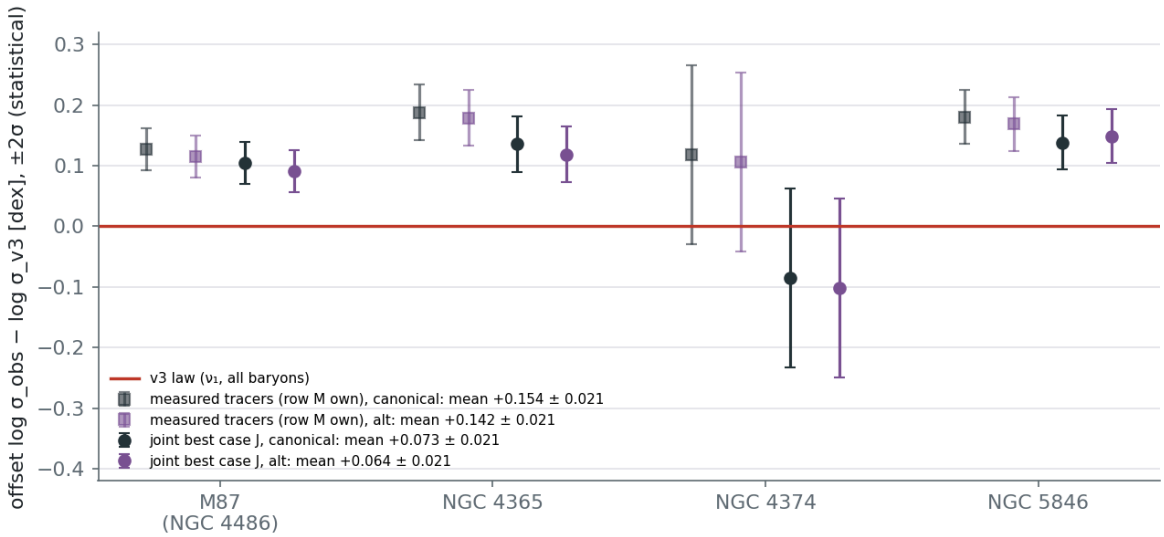
Central ellipticals (SLUGGS)

NOT DECIDABLE YET

On the board: tension 2.8σ → 3.1σ (3.3σ without ownership)

Central ellipticals (SLUGGS globular-cluster kinematics)

UNDECIDED: genuine tension 3.1σ in the joint best case (3.3σ without ownership); 1 of 4 cleared



source: rebuild_2026_10_22/S4_v3_muK_clusters/r4_out/cfg528b_measured_tracers/{nu1,nu1_noown}

ON THE PLATE

Globular-cluster velocity dispersions at large radius around four central ellipticals (M87, NGC 4365, NGC 4374, NGC 5846) from SLUGGS, with measured GC density profiles and anisotropy where available (CFG528b), rerun by S4 with v3's kernel ν_1.

HOW IT IS PLATED

Each galaxy's offset between the measured and the v3-predicted dispersion (dex) with ±2σ statistical bars, for the measured-tracer row and for the joint best case (every admissible lever pushed in the law's favour), on both footings. The red line at zero is the law.

WHY IT IS NOT DECIDABLE YET

Three of the four centrals sit well above the law, by 0.09–0.15 dex even in the joint best case; only NGC 4374 is cleared. The class tension is 3.1σ on the less favourable footing (3.3σ if host ownership is dropped, as v3 has no ownership rule). Mass-side levers (IMF, distance, hot gas, external field) cannot remove it; only the unmeasured tracer slope and anisotropy for three of the four could, so the tile stays undecided. The errors are statistical only.

Sources:
rebuild_2026_10_22/S4_v3_muK_clusters/r4_out/cfg528b_measured_tracers/nu1/campaign_fresh_gravity__CFG528_sluggs_centrals_mechanism__cfg528b_measured_tracers_results.json @ 330b00686a;
rebuild_2026_10_22/S4_v3_muK_clusters/r4_out/cfg528b_measured_tracers/nu1_noown/campaign_fresh_gravity__CFG528_sluggs_centrals_mechanism__cfg528b_measured_tracers_results.json @ 330b00686a;
campaign_fresh_gravity/CFG528_sluggs_centrals_mechanism/cfg528b_measured_tracers.out @ 63f7fea1b3;
rebuild_2026_10_22/S4_v3_muK_clusters/README.md @ 330b00686a

Lanes: CFG528b / S4

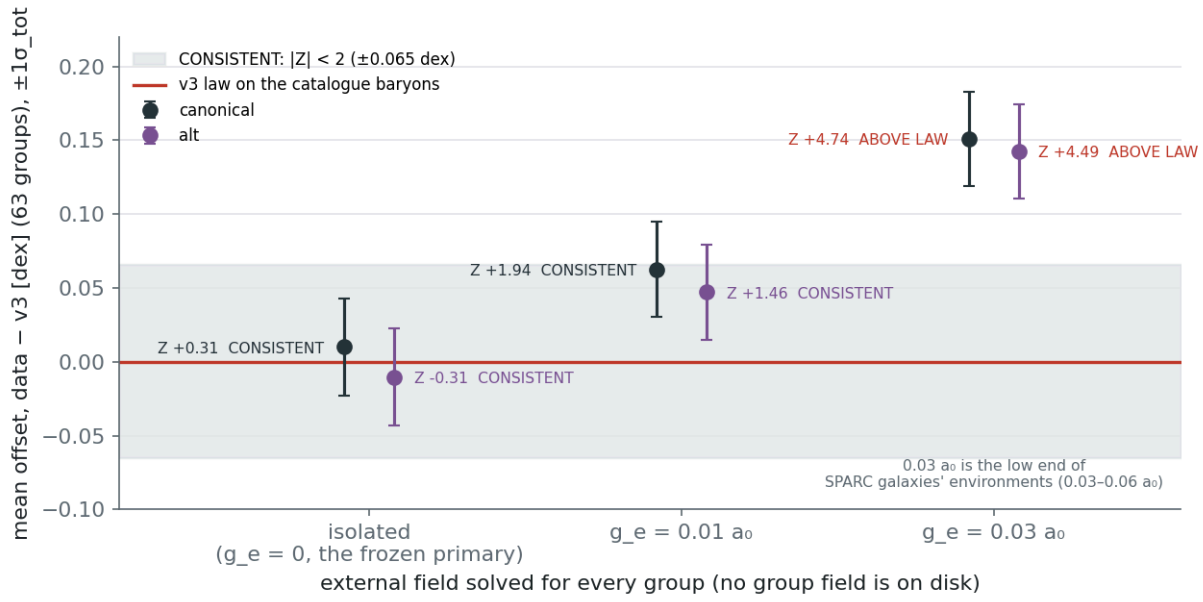
Galaxy groups (BFJR)

PASSES

On the board: isolated: +0.010/−0.010 dex (Z ±0.31), no supply rule; ABOVE LAW if g_e ≥ 0.03 a0 (Z 4.7/4.5); no group field on disk

Galaxy groups: the baryonic Faber–Jackson relation

PASS (isolated): +0.010 / -0.010 dex; ABOVE LAW if the groups sit in a field ≥ 0.03 a_0



source: rebuild_2026_10_22/P4_v3_galaxy_tiles/p4_v3_galaxy_tiles_results.json tiles.T3.footings

ON THE PLATE 63 galaxy groups from Tian+2026 (member velocity dispersion, mean projected radius, K-band + X-ray-gas baryonic mass), scored with v3's law on the catalogue baryons with a whole-system aperture.

HOW IT IS PLATED The mean offset of the groups from v3 (with total error) when the groups are taken as isolated, and when an external field of 0.01 and 0.03 a_0 is solved, on both footings. The grey band is the CONSISTENT zone |Z| < 2.

WHY IT PASSES Taken as isolated, the groups sit on the law: +0.010 / −0.010 dex, Z ±0.31, with no supply or edge rule (v3 has none). Every aperture variant stays consistent. The pass depends on the groups' external field, which is not on disk: at 0.01 a_0 they stay consistent (Z 1.94 / 1.46), but at 0.03 a_0, the low end of SPARC environments, they sit above the law (Z 4.7 / 4.5). The baryon masses inherit a Kroupa K-band M/L. The μ_K^2 term was not included.

Sources: rebuild_2026_10_22/P4_v3_galaxy_tiles/p4_v3_galaxy_tiles_results.json @ 6c0249a8bc; rebuild_2026_10_22/P4_v3_galaxy_tiles/README.md @ 6c0249a8bc

Lanes: P4

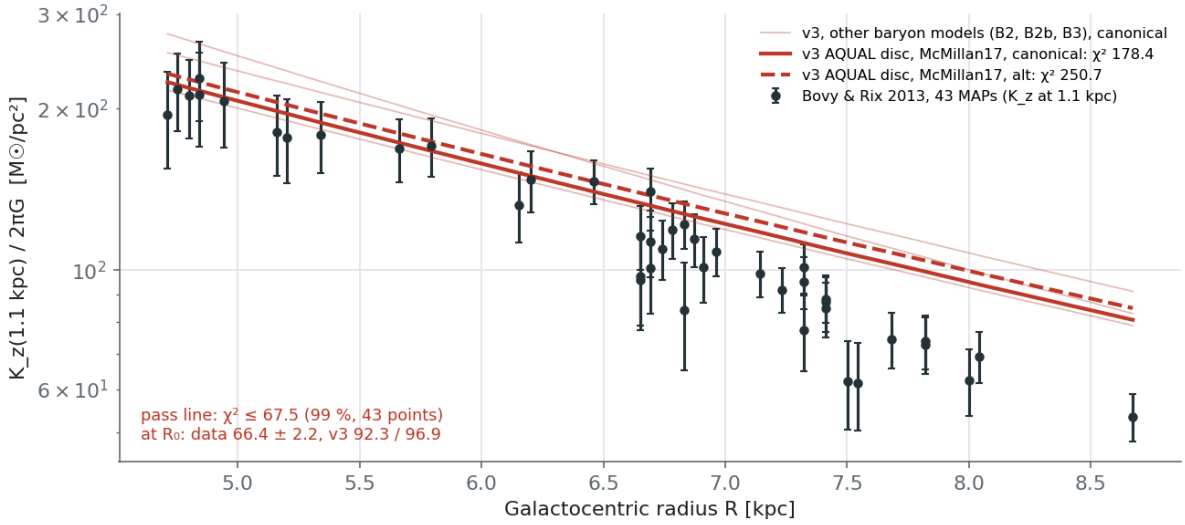
MW vertical force (round rule)

FAILS

On the board: v3 has no round rule; AQUAL disc K_z,1.1 92.3/96.9 vs 66.4 ± 2.2, χ^2 178/251 vs 67.5; all four baryon models

Milky Way vertical force

FAIL: v3's disc-like AQUAL phantom over-predicts K_z for every baryon model on both footings



source: rebuild_2026_10_22/P4_v3_galaxy_tiles/p4_v3_galaxy_tiles_results.json tiles.T4.models; real_research/data/mw_kz11_bovyrix2013_table3.tsv

ON THE PLATE The vertical force K_z at 1.1 kpc above the Galactic plane for 43 mono-abundance populations (Bovy & Rix 2013, with a 5 % error floor), against v3's AQUAL solution for the Milky Way's baryons (McMillan17 and three bracketing baryon models).

HOW IT IS PLATED Measured K_z against radius (log scale) with errors; the thick red lines are v3 on the canonical (solid) and alt (dashed) footings for McMillan17's baryons, the faint lines are v3 for the other baryon models.

WHY IT FAILS The record's pass used a round enclosed-mass rule that v3 does not have. v3's own AQUAL phantom is disc-like, so it adds vertical force where the disc is: K_z at the Sun comes out 92.3 / 96.9 against 66.4 ± 2.2, with a longer scale length than the data. χ^2 is 178 / 251 against the 67.5 line, and every baryon model fails, as do screening and a 0.03 a_0 field. McMillan17 is partly halo-fitted; the other models bracket that.

Sources: rebuild_2026_10_22/P4_v3_galaxy_tiles/p4_v3_galaxy_tiles_results.json @ 6c0249a8bc; rebuild_2026_10_22/P4_v3_galaxy_tiles/README.md @ 6c0249a8bc; real_research/data/mw_kz11_bovyrix2013_table3.tsv @ f33d4e86a9

Lanes: P4

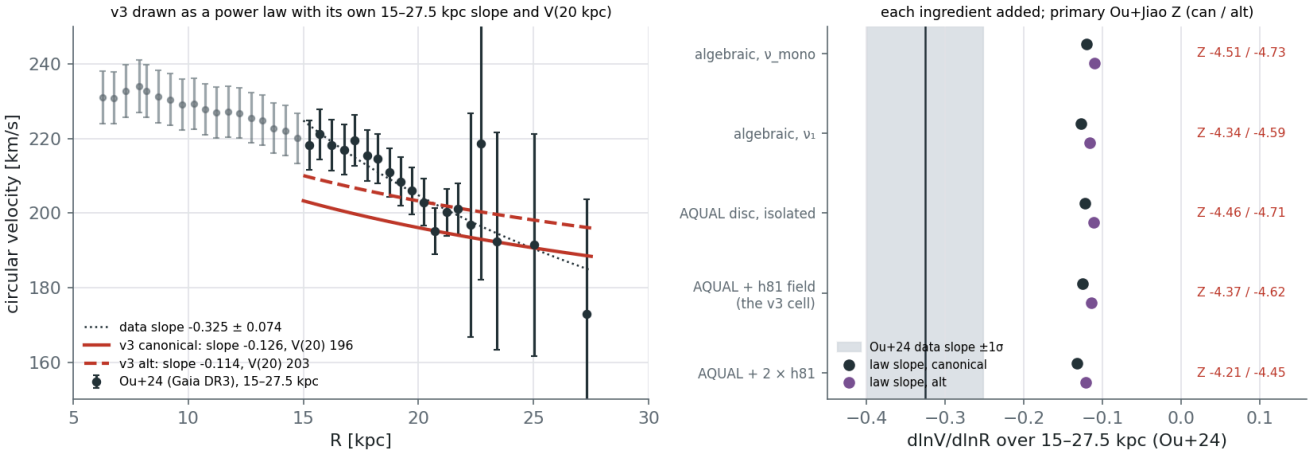
MW outer slope (Gaia)

FAILS

On the board: SHAPE EXCLUDED: law slope −0.126/−0.114 vs data −0.325, Z −4.37/−4.62; ALG→ν_1→AQUAL disc→EFE all fail

Milky Way outer rotation-curve slope (Gaia)

FAIL (shape excluded): v3's slope -0.126 / -0.114 vs data -0.325; Z -4.37 / -4.62



source: rebuild_2026_10_22/S3_v3_mw_seals/s3_partA_mw_results.json steps; real_research/data/mw_rc_ou2024_table1.tsv

ON THE PLATE

The Milky Way's circular velocity from Gaia DR3 red giants (Ou+24, 37 points; Jiao+23 in the primary test) over 15–27.5 kpc, against v3's AQUAL disc solution for McMillan17's baryons in the Galaxy's external field.

HOW IT IS PLATED

Left: the measured rotation curve (window points dark, others grey) with the data's slope (dotted) and v3 drawn as a power law with its own fitted slope and V(20 kpc), on both footings. Right: v3's slope as each ingredient is added (kernel, disc geometry, external field), against the data slope band, with the primary Z.

WHY IT FAILS

The Gaia curve falls steeply beyond 15 kpc (slope −0.325); v3 falls only gently (−0.126 / −0.114), so the shape is excluded at Z −4.37 / −4.62. No ingredient cures it: the kernel, the AQUAL disc solve and the external field each move the slope by about 0.01, and the μ_K^2 term (first-order bound) makes it shallower still. The cause is the steep DR3 decline itself, not the law's amplitude: at the census baryons v3 also runs low (V(20 kpc) 196.1 / 203.3 km/s), but rescaling the stellar mass to fix the amplitude leaves the slope at -0.147 / -0.126. Cepheids (Feng+26) are shape-consistent. The power-law lines on the left are a drawing aid; the verdict uses the lane's solved curves.

Sources: rebuild_2026_10_22/S3_v3_mw_seals/s3_partA_mw_results.json @ f1b4c8c687; rebuild_2026_10_22/S3_v3_mw_seals/README.md @ a2e4a36598; real_research/data/mw_rc_ou2024_table1.tsv @ f33d4e86a9

Lanes: S3

~ House specials ~

The excess family, still open · 5 dishes

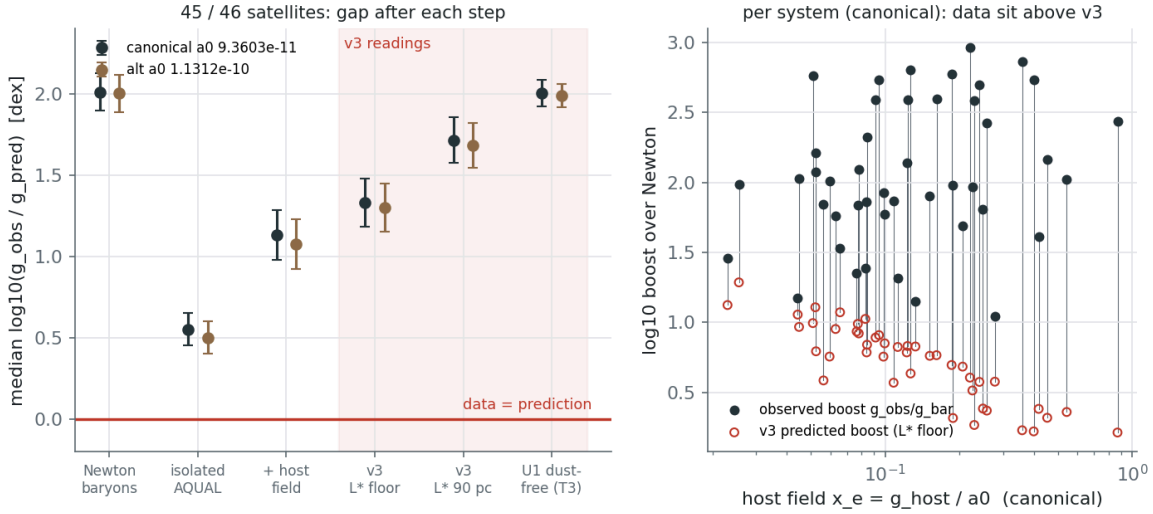
Dwarf satellites vs screened law

FAILS

On the board: v3, host EFE solved: +1.33/+1.30 dex (L★ floor), +1.72/+1.68 at 90 pc; supply (P1), no binding; U1: the theorem (T3) that passes the tidal dwarfs leaves them dust-free: +2.00/+1.99 dex; STRUCTURAL

Dwarf satellites vs the screened law

FAIL: +1.33 / +1.30 dex above v3 at the L* floor (9.0 / 8.8 sd); dust-free (U1) +2.00 / +1.99



source: S1_v3_hosted_small (results json, s1_per_system.csv); U1_v3_dust_content N5

ON THE PLATE E3's clean primary sample of Milky Way / M31 dwarf satellites (45 canonical, 46 alt systems; Arroyo free-f binaries, Upsilon_V = 2, E-W estimator). Statistic: bootstrap median over systems of log10(g_obs/g_pred). v3 = stiffened AQUAL mu = x/(1+x) solved for each satellite as a Plummer sphere in its host's uniform field.

HOW IT IS PLATED Left: the median gap (dex) after each modelling step, both footings with bootstrap sd; the red line at 0 is 'data = prediction'. The shaded steps are v3 readings (L* floor, L* 90 pc) and U1's dust-free hosted case. Right: per satellite (canonical) the observed boost over Newton (dark dots) and v3's predicted boost (open red circles) against the host field x_e, joined by grey lines.

WHY IT FAILS Isolated, the law leaves the satellites about +0.55 dex short; putting them in their host's field (the EFE that v3's AQUAL has) pulls the prediction toward Newton and the gap grows to +1.33 / +1.30 dex at the L* floor (9.0 / 8.8 sd), more at L* = 90 pc. The frozen FAIL line (gap >= 3 sd) is met on both footings, also for the worst-clean data. U1's theorem T3 says a hosted system that falls with its host carries no held dust, so the satellites would be dust-free and sit +2.00 / +1.99 dex above: STRUCTURAL. Each satellite is a spherical Plummer in a uniform field (no tides, no non-equilibrium).

Sources: rebuild_2026_10_22/S1_v3_hosted_small/s1_v3_hosted_small_results.json @ 2b6bb658a1; rebuild_2026_10_22/S1_v3_hosted_small/s1_per_system.csv @ 2b6bb658a1; rebuild_2026_10_22/U1_v3_dust_content/u1_v3_dust_content_results.json @ bbb6bfcc71;

rebuild_2026_10_22/S1_v3_hosted_small/README.md @ 2b6bb658a1;
rebuild_2026_10_22/U1_v3_dust_content/README.md @ 58de634d74

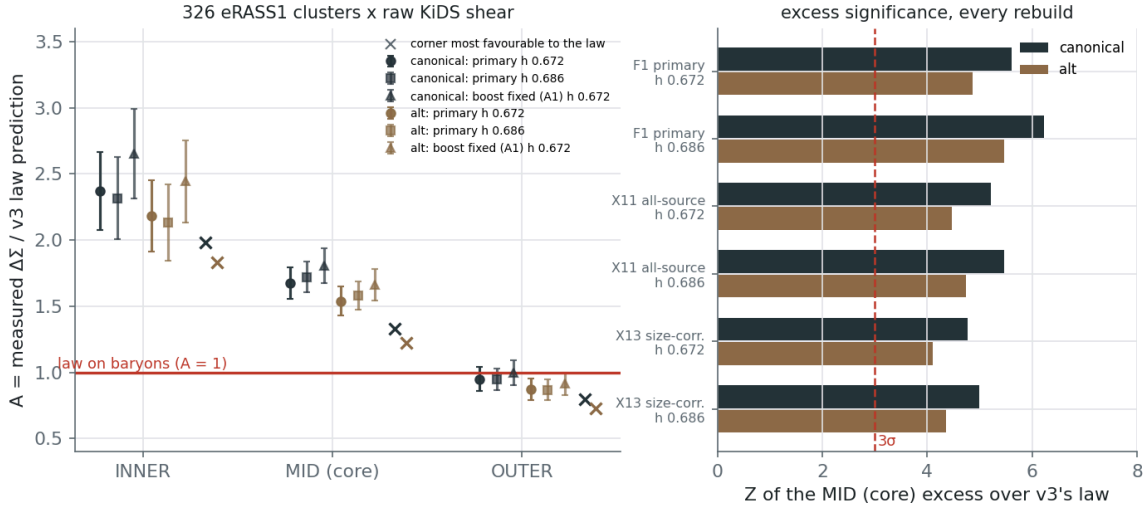
Lanes: S1 / P1 / U1

Cluster cores (326 eRASS1) FAILS

On the board: x1.68-1.85 / x1.54-1.70 the law (F1), Z 4.5-5.5 (X11), 4.1-5.0 size-corrected (X13: the size trend was a shear selection of the split); no μ_K fixes it (S4); MOG fails too (X9)

Cluster cores (326 eRASS1)

FAIL: core shear x1.68 / x1.54 the law (h 0.672); Z 4.1-6.2 across F1, X11, X13



source: F1_cluster_cores, X11_cross_shear_diagnosis, X13_source_size_test (results json)

ON THE PLATE

Stacked tangential shear (ΔΣ) of 326 eRASS1 clusters from raw KiDS sources (21.3 million rows), in INNER / MID / OUTER radial apertures. v3 prediction: AQUAL mu = x/(1+x) on the clusters' baryons (eRASS1 gas, stars), native h 0.672 and 0.686, catalogue miscentring. A = measured / predicted.

HOW IT IS PLATED

Left: A in the three apertures for both footings (dark = canonical, brown = alt), three primary cells each with error bars; crosses are the corner most favourable to the law; red line A = 1 is the law's prediction. Right: the Z of the MID (core) excess in every rebuild (F1, X11 all sources, X13 corrected for the size-split selection), with the 3σ line.

WHY IT FAILS

The core (MID) shear sits 1.5–1.9 times above what the law gives on the clusters' baryons, at Z 4.1–6.2 on every rebuild and both footings, while OUTER is close to 1: a shape problem, not an amplitude knob. Even the corner most favourable to the law (stars f* 0.30, gas +1σ, deep-X-ray gas factor, X-COP shape) leaves MID 1.23–1.37 and pushes OUTER below 1. X13 showed X11's source-size trend was a shear selection of the split, so the excess stands. S4: no allowed μ_K (v3's dust mass term) fixes it; X9: MOG fails too. Inherited inputs: eRASS1 R500 and distances use flat ΛCDM (70, 0.3), bracketed in F1.

Sources: rebuild_2026_10_22/F1_cluster_cores/f1_cluster_cores_results.json @ f1803b0256;
rebuild_2026_10_22/X11_cross_shear_diagnosis/x11_analysis_results.json @ c1cd5455a5;
rebuild_2026_10_22/X13_source_size_test/x13_analysis_results.json @ 4eeeceba07;
rebuild_2026_10_22/F1_cluster_cores/README.md @ f1803b0256; rebuild_2026_10_22/S4_v3_muK_clusters/README.md @ 330b00686a; rebuild_2026_10_22/X9_mog_stvg_check/README.md @ ef0a3d7fee;
rebuild_2026_10_22/X13_source_size_test/README.md @ 12b3f8bcba

Lanes: CFG687 / F1 / S4 / X7 / X9 / X11 / X13

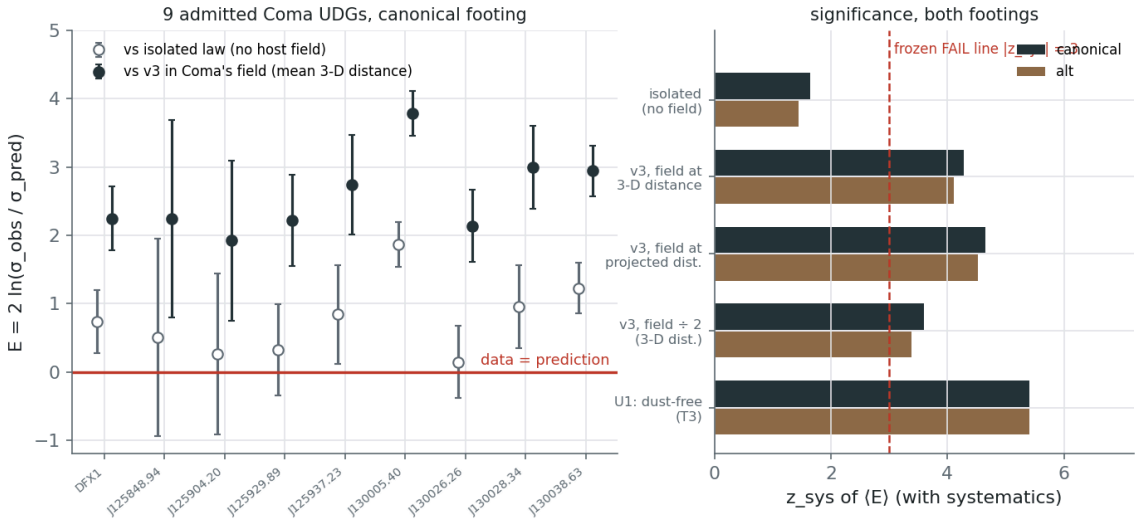
Coma ultra-diffuse galaxies

FAILS

On the board: in Coma's field +1.12/+1.08 dex, z +4.3–4.7 / +4.1–4.5; isolated only 1.65/1.45σ off; U1: +1.41 dex, z 5.4 under v3 dynamics; STRUCTURAL

Coma ultra-diffuse galaxies

FAIL: in Coma's field +1.12 / +1.08 dex above v3, z_sys 4.3 / 4.1; isolated only 1.65 / 1.45



source: S1_v3_hosted_small (results json, s1_coma_per_galaxy.csv); U1_v3_dust_content N5

ON THE PLATE

CFG933's 9 admitted Coma ultra-diffuse galaxies (stellar velocity dispersions σ_obs, M★, R_e). Statistic E = 2 ln(σ_obs/σ_pred), unweighted mean ⟨E⟩, systematics from CFG948's recipe rerun on the v3 prediction. v3: stiffened AQUAL solved in Coma's measured hydrostatic field (β-model) at the mean 3-D and projected distances.

HOW IT IS PLATED

Left: E per galaxy (canonical) with error bars, against the isolated law (open grey) and against v3 in Coma's field (dark); red line E = 0 is the prediction. Right: z_sys of ⟨E⟩ for each reading, both footings, with the frozen FAIL line at 3.

WHY IT FAILS

Isolated, the UDGs sit only 1.6 / 1.4σ above the law. But v3 has an external-field effect, and Coma's measured field (x_e ≈ 0.50–1.06 a0) pushes the prediction toward Newton, so all 9 sit above it: ⟨E⟩ +2.58 / +2.48 ln (+1.12 / +1.08 dex), z_sys 4.3 / 4.1, and still 3.4–3.6 with the field halved. Screening is irrelevant here. U1's theorem T3 (held dust is weightless in its host) makes it worse: +1.41 dex, z_sys 5.4. The fail assumes equilibrium in the current field; a UDG far out of equilibrium would escape it.

Sources: rebuild_2026_10_22/S1_v3_hosted_small/s1_v3_hosted_small_results.json @ 2b6bb658a1;
rebuild_2026_10_22/S1_v3_hosted_small/s1_coma_per_galaxy.csv @ 2b6bb658a1;
rebuild_2026_10_22/U1_v3_dust_content/u1_v3_dust_content_results.json @ bbb6bfcc71;
rebuild_2026_10_22/S1_v3_hosted_small/README.md @ 2b6bb658a1

Lanes: S1 / U1

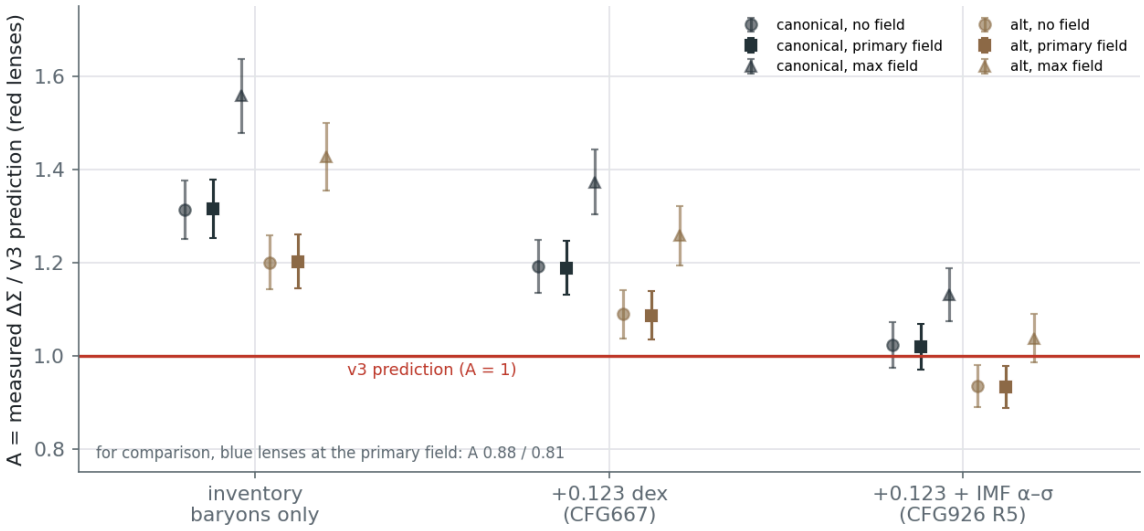
Red (early-type) lenses

CONDITIONAL

On the board: inventory 1.31/1.20 short; + CFG667 + IMF α–σ → 1.02/0.93

Red (early-type) lenses

CONDITIONAL: inventory 1.31 / 1.20 short; with CFG667 + IMF α–σ 1.02 / 0.93 (consistent)



source: S2_v3_lowg_lensing/s2_v3_lowg_lensing_results.json (cells_brackets)

ON THE PLATE

KiDS galaxy-galaxy lensing of the record's 181,477 isolated lenses (B21 points in CFG829's low-g window, g_bar 10^-13.3 to 10^-11.9 m/s², 6 points), split by colour. v3 is forward-modelled as 4GΔΣ of F2's AQUAL solution with inventory baryons, hot CGM 0.5 M★, and the host field at three levels. A = data / v3.

HOW IT IS PLATED

Red-lens amplitude A for three baryon readings (inventory only; +0.123 dex from CFG667; +0.123 dex plus the IMF α–σ posit), at three host-field levels, both footings, with error bars. Red line A = 1 is v3. The blue-lens A at the primary field is quoted for comparison.

WHY IT IS CONDITIONAL

On the stellar inventory alone, red lenses lens 1.32 / 1.20 times more than v3 predicts (5.0σ canonical, 3.5σ alt by the README), while blue lenses are consistent. The excess is removed only by two declared posits: CFG667's +0.123 dex extra baryons and a bottom-heavy IMF that scales with σ (CFG926), which bring A to 1.02 / 0.93. So the tile is conditional on those posits; with M★ −0.2 dex the red tile fails on canonical even with the IMF. The external field barely matters in this window except at the maximum field. The μ_K²ψ term is not included.

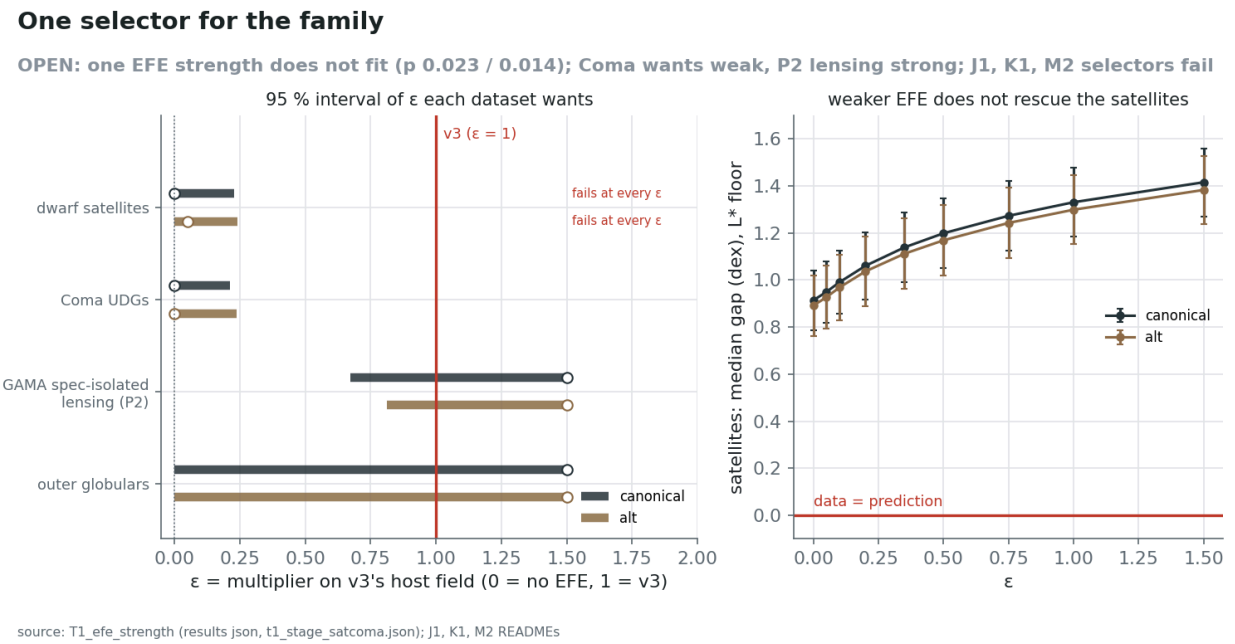
Sources: rebuild_2026_10_22/S2_v3_lowg_lensing/s2_v3_lowg_lensing_results.json @ 0a3db18162;
rebuild_2026_10_22/S2_v3_lowg_lensing/README.md @ 0a3db18162;
rebuild_2026_10_22/S2_v3_lowg_lensing/FROZEN_CRITERIA.md @ c94150c05d

Lanes: S2 / CFG667 / CFG926

One selector for the family

OPEN

On the board: J1, K1, M2 selectors fail; T1: one EFE strength does not fit (p 0.023/0.014); Coma wants weak, P2 lensing strong; satellites +0.9 dex even at ε = 0



source: T1_efe_strength (results json, t1_stage_satcoma.json); J1, K1, M2 READMEs

ON THE PLATE

T1 scales v3's host field by one diagnostic number ε (ε = 1 is v3, ε = 0 no external-field effect) and re-runs each member through its own lane's solver: S1's satellites and Coma UDGs, P2's GAMA spectroscopic isolated-galaxy lensing, D10's outer globulars. Selector lanes J1 (potential depth), K1 (braneworld Weyl fluid) and M2 (K1 + L* screening) are cited from their READMEs.

HOW IT IS PLATED

Left: each dataset's 95 % interval in ε, both footings (open circle = best ε; 1.5 is the scan edge); the red line is v3 at ε = 1. Right: the satellites' median gap against ε for both footings; it never reaches the red zero line.

WHY IT IS STILL OPEN

A single selector would have to explain why satellites, Coma UDGs and cluster cores sit above v3 while other systems do not. Measured as one EFE strength, the data disagree with each other: Coma wants ε ≤ 0.21 / 0.24, P2 lensing wants ε ≥ 0.67 / 0.81, and the satellites fail at every ε (still +0.91 / +0.89 dex at ε = 0). The one-ε consistency test gives p = 0.023 / 0.014, just above the frozen INCONSISTENT line (0.01). The potential-depth selector J1, the braneworld fluid K1 and its screened version M2 all fail their frozen rules. The family's membership still has to be re-derived after R1/R2/R6, so the tile stays open.

Sources: rebuild_2026_10_22/T1_efe_strength/t1_efe_strength_results.json @ 31e3c85bf6;
rebuild_2026_10_22/T1_efe_strength/t1_stage_satcoma.json @ 31e3c85bf6;
rebuild_2026_10_22/T1_efe_strength/README.md @ 31e3c85bf6;
rebuild_2026_10_22/J1_potential_selector/README.md @ 3d75573e2c;
rebuild_2026_10_22/K1_braneworld_weyl/README.md @ 83d3592c87;
rebuild_2026_10_22/M2_cluster_compact/README.md @ d5f6495142

Lanes: CFG914 / CFG919 / CFG921 / CFG928 / J1 / K1 / M2 / T1

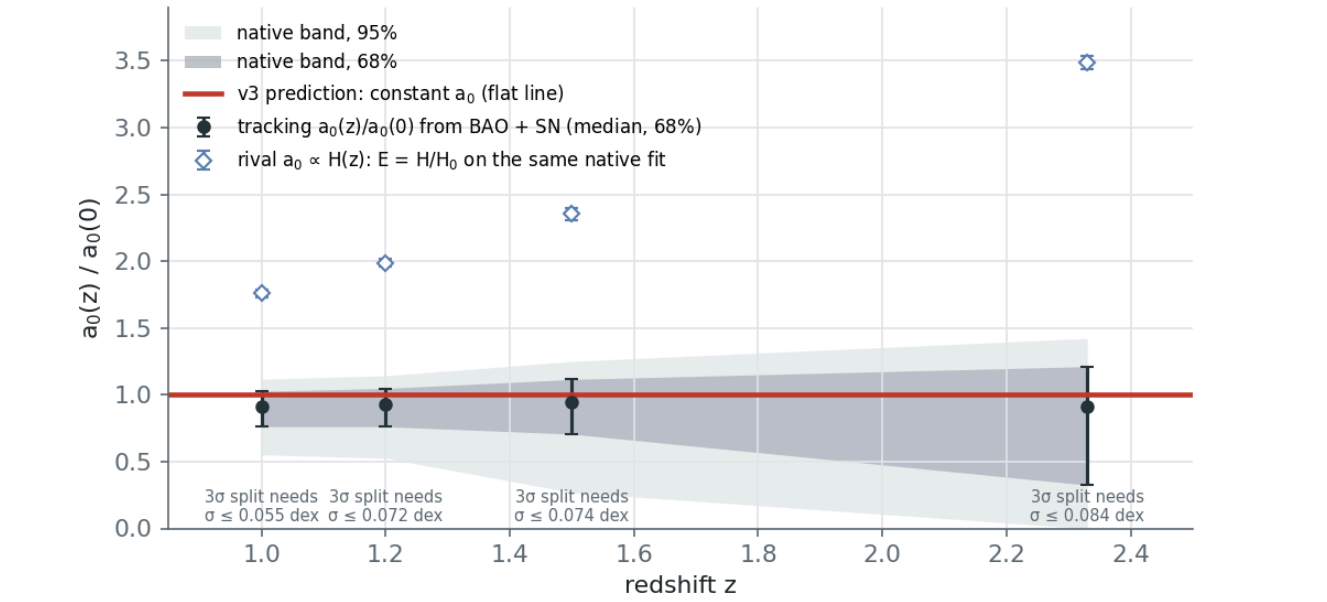
~ From the cellar ~

Cosmology and a0 over time · 9 dishes

a0(z): native band vs a0 ~ H(z)

NOT DECIDABLE YET

On the board: own BAO ruler band; data cannot decide yet



source: campaign_fresh_gravity/CFG689_native_ruler_bao/cfg689.json (primary: curvature free, DESI DR2 BAO + Pantheon+)

ON THE PLATE

CFG689 primary mode: DESI DR2 anisotropic BAO (LRG1, LRG2, LRG3+ELG1, ELG2, QSO, Lya) plus BGS D_V, and 1580 Pantheon+ supernovae (zHD > 0.01), on the framework's own sound-horizon ruler (r_d0 = 147.07 Mpc), curvature free. Dark energy is free in six nodes; the law's tracking reading a0(z)/a0(0) = sqrt(rho_DE(z)/rho_DE(0)) is read off at z = 1.0, 1.2, 1.5, 2.33.

HOW IT IS PLATED

x: redshift. y: a0(z)/a0(0). Grey bands: the 68% and 95% range of the tracking reading from the expansion data; black points: its median with 68% bars. Red line: v3's own prediction, a constant a0 (ratio 1). Open blue diamonds: the rival a0 proportional to H(z), i.e. E = H/H0 from the same fit. Small grey text under each node: the galaxy-side a0 error that would be needed to split the two at 3 sigma.

WHY IT IS NOT DECIDABLE YET

v3's action has a constant a0, so it predicts the flat line, and the flat line lies inside the expansion data's band at every node. The rival sits well above the band (closest gap +0.23 to +0.45 dex), so the expansion side has already separated the two readings (CFG689: DEGENERACY BROKEN). The decision now rests on kinematic a0 measurements of galaxies at those redshifts, and none of the record's samples reaches the needed 0.055 to 0.084 dex (CFG689 grades KURVS, the z~1.2 multi-tracer set and realistic RC100/CRISTAL as

CANNOT; only the optimistic RC100/CRISTAL case LEANS). Inherited input: the omega_b and omega_c priors (0.02236, 0.1202) are CMB-fit values; CFG689's B5 brackets without them weaken the separation (PARTLY BROKEN). The ratio is the same on both footings; the absolute a0 is not drawn.

Sources: campaign_fresh_gravity/CFG689_native_ruler_bao/cfg689.json @ 6c064b83a3; campaign_fresh_gravity/CFG689_native_ruler_bao/cfg689_summary.md @ 6c064b83a3

Lanes: CFG689 / PAPER38

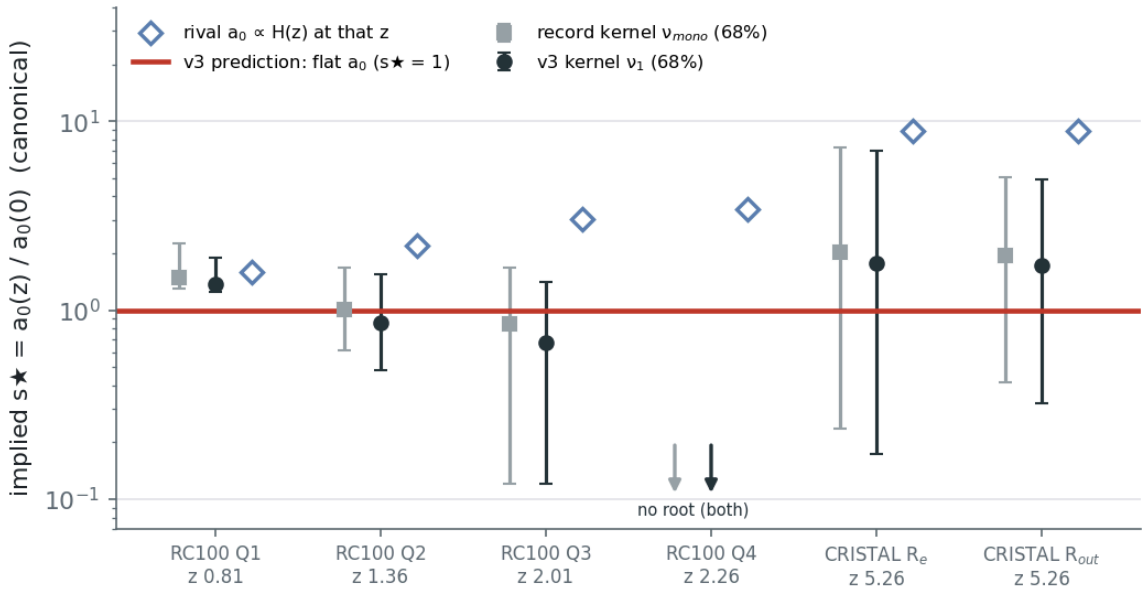
High-z on halo-free inputs

NOT DECIDABLE YET

On the board: all six CFG303 classes unchanged; s★ ≤ 0.1 dex

High-z on halo-free inputs

v3's kernel moves s★ by at most 0.1 dex; every class unchanged; flat vs H(z) not separated



source: rebuild_2026_10_22/S4_v3_muK_clusters/r4_out/cfg303_rc100_cristal_LCDMFREE/{repro,nu1}/..._results.json (S4 R4 swap of CFG303)

ON THE PLATE	CFG303's halo-free high-z points: RC100 sample B (100 galaxies, SED stellar mass from Table 3 col. 6 plus scaling gas, thin disc at R_e) in four redshift quartiles, and the six CRISTAL discs at z ~ 5.26 at R_e and at the outermost data marker. Each point is recomputed by S4's R4 wrapper twice: with the record's kernel nu_mono and with v3's kernel nu_1 = (1 + sqrt(1 + 4/y))/2.
HOW IT IS PLATED	x: one column per point (name and median z). y (log): implied s* = a0(z)/a0(0) on the canonical footing, with 68% bars. Grey squares: record kernel; black dots: v3 kernel. Red line: v3's flat a0 (s* = 1). Open blue diamonds: what a0 proportional to H(z) expects at that z. Arrows: no root (the native baryons already meet or exceed the dynamics).
WHY IT IS NOT DECIDABLE YET	Swapping in v3's kernel moves s* down by at most about 0.1 dex (RC100 Q3: 0.84 to 0.67), well inside each point's spread, and S4 finds every CFG303 class unchanged. Flat a0 lies inside the 95% interval of every rooted point; the rival's expectation falls outside it at RC100 Q3 and CRISTAL R_out (8.84 against an upper edge of 8.64), and RC100 Q4 has no root under either kernel, which neither law explains. CFG303 reads all of this as calibration-limited: the shifts come from 0.05 to 0.10 dex of baryon mass and geometry acting through a lever of -2.5 to -4.8 dex per dex, so none of it separates flat a0 from a0 proportional to H(z). The velocities are still the authors' fitted V_c (MODEL-OTHER, halo in the fit), and the rival's expectation uses CFG223's flat-Lambda E(z) background, an inherited input. The record kernel's rc100 script failed its reproduction check in 5 of 1,667 leaves (record drift in the MNRAS S5 block, not in these points).

Sources:
rebuild_2026_10_22/S4_v3_muK_clusters/r4_out/cfg303_rc100_cristal_LCDMFREE/nu1/campaign_fresh_gravity__CFG303_lcdm_free_inputs__cfg303_rc100_cristal_LCDMFREE_results.json @ 330b00686a;
rebuild_2026_10_22/S4_v3_muK_clusters/r4_out/cfg303_rc100_cristal_LCDMFREE/repro/campaign_fresh_gravity__CFG303_lcdm_free_inputs__cfg303_rc100_cristal_LCDMFREE_results.json @ 330b00686a;
rebuild_2026_10_22/S4_v3_muK_clusters/README.md @ 330b00686a;
campaign_fresh_gravity/CFG303_lcdm_free_inputs/README.md @ 2d9bdc1b92

Lanes: CFG303 / S4

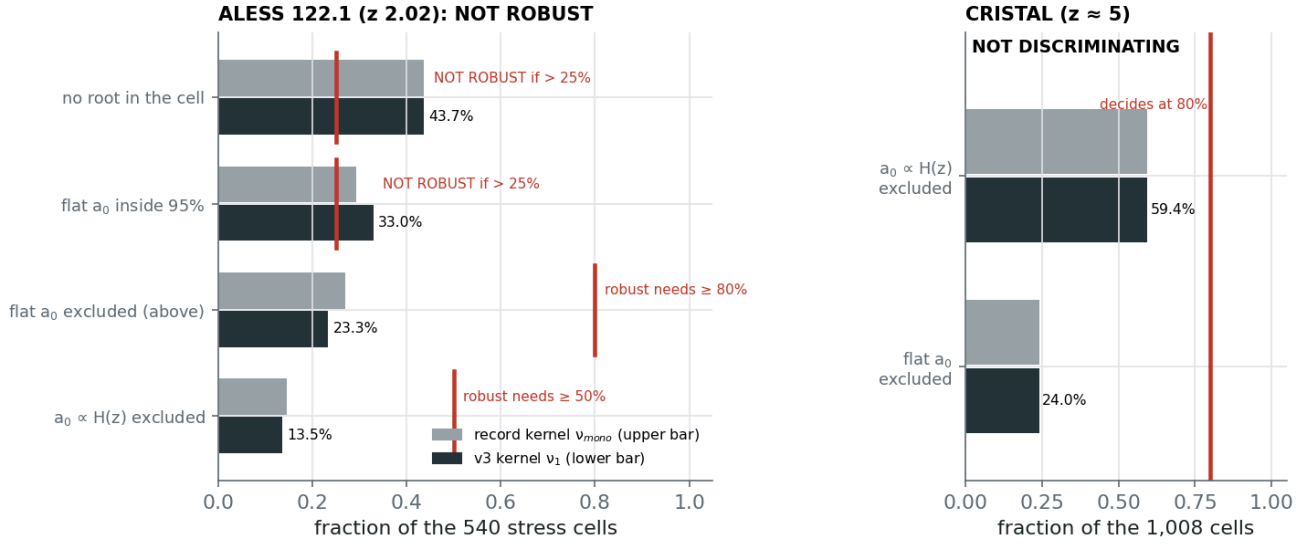
CRISTAL z~5 / ALESS 122.1

NOT DECIDABLE YET

On the board: NOT ROBUST / NOT DISCRIMINATING *unchanged*

CRISTAL z~5 / ALESS 122.1

ALESS trips both NOT ROBUST triggers; CRISTAL excludes neither law in 80% of cells; v3 kernel changes no class



source: rebuild_2026_10_22/S4_v3_muK_clusters/r4_out/cfg307_stress/*, cfg308_cristal_stress/* (repro = record, nu1 = v3); CFG307/CFG308 frozen rules

ON THE PLATE

Two stress tests on framework-native inputs. CFG307: ALESS 122.1 (z 2.024) over a 540-cell grid of the record's own choices (gas tracer, radius, pressure term, inclination, M*). CFG308: the CRISTAL z ~ 5 discs over 1,008 cells. Both re-run by S4's R4 wrapper with the record kernel and with v3's nu_1.

HOW IT IS PLATED

Left: ALESS 122.1, four horizontal bar pairs (upper grey = record kernel, lower black = v3 kernel): fraction of cells with no root, with flat a0 inside the 95% interval, with flat a0 excluded, with a0 proportional to H(z) excluded. Red ticks: the frozen decision lines (25% triggers for NOT ROBUST; 80% and 50% for a robust exclusion). Right: CRISTAL, fraction of cells excluding H(z) and excluding flat a0, against the frozen 80% line.

WHY IT IS NOT DECIDABLE YET

For ALESS 122.1 both NOT ROBUST triggers fire under both kernels: 43.7% of cells have no root and flat a0 sits inside the interval in 33.0% (v3), while flat is excluded in only 23.3% and H(z) in 13.5%, far from the 80% and 50% a robust exclusion needs. For CRISTAL, H(z) is excluded in 59.4% of cells and flat in 24.0% under both kernels, and neither reaches the 80% line, so the point is NOT DISCRIMINATING. v3's kernel changes no class. Neither point is an a0(z) measurement; both are limited by gas and pressure choices.

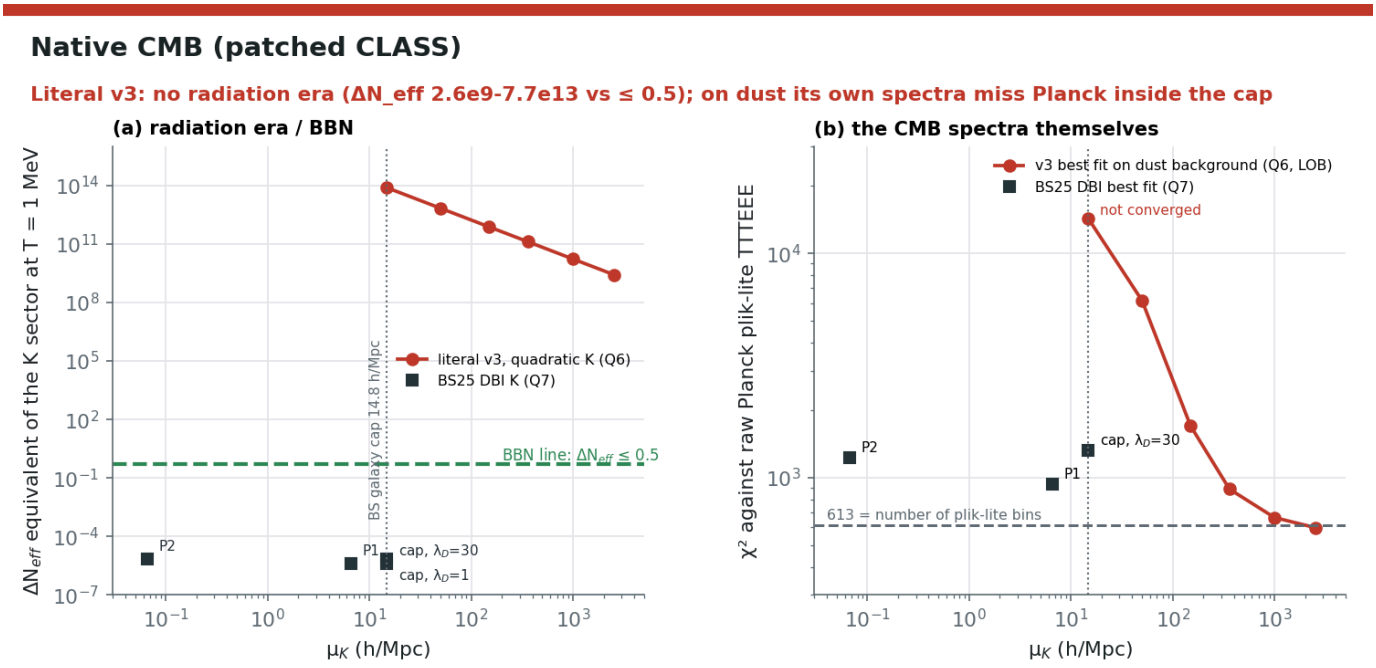
Sources:
rebuild_2026_10_22/S4_v3_muK_clusters/r4_out/cfg307_stress/nu1/campaign_fresh_gravity__CFG307_aless122_stress_test__cfg307_stress_results.json @ 330b00686a;
rebuild_2026_10_22/S4_v3_muK_clusters/r4_out/cfg308_cristal_stress/nu1/campaign_fresh_gravity__CFG308_cristal_stress_test__cfg308_cristal_stress_results.json @ 330b00686a;
campaign_fresh_gravity/CFG307_aless122_stress_test/README.md @ a4a6d3c1ad;
campaign_fresh_gravity/CFG308_cristal_stress_test/README.md @ bc2038fff0

Lanes: CFG307 / CFG308 / S4

Native CMB (patched CLASS)

FAILS

On the board: literal v3: no radiation era (stiff a^-6 term of BS's K); on dust, plik-lite $\Delta\chi^2$ vs ΛCDM +13891 at the cap, +18.7 at μ_K 2542; Q7: BS25's DBI K fixes BBN only, R_B 0.26–0.32; BS ran no Boltzmann code



source: rebuild_2026_10_22/Q6_v3_class/q6_native_results.json, rebuild_2026_10_22/Q7_bs25_cosmology/q7_native_results.json (plik-lite bin count: q6_native.py)

ON THE PLATE

Q6: v3's linear equations derived from the action and coded into CLASS v3.3.4 (khronon patch), at mu_K = 14.8 to 2542 h/Mpc. Literal v3 (quadratic K) background, and fits on v3's dust background with the literal perturbations (LOB) to raw Planck plik-lite TTTEEE (613 bins) plus low-l TT and EE. Q7: the same with Blanchet-Skordis 2025's DBI K at their parameter sets P1, P2 and at the cap (lambda_D = 1, 30).

HOW IT IS PLATED

Left (log-log): the K sector's energy at T = 1 MeV, as an equivalent Delta N_eff, against mu_K; red: literal v3; black squares: DBI K. Green dashed: the BBN requirement Delta N_eff <= 0.5. Right (log-log): chi2 of v3's own best fit against the raw plik-lite spectra, against mu_K; grey dashed: 613, the number of plik-lite bins, as a scale. Dotted vertical line: BS's galaxy cap, 14.8 h/Mpc.

WHY IT FAILS

Literal v3 has no radiation era: its stiff a^-6 term carries 2.6e9 to 7.7e13 Delta N_eff at 1 MeV across the grid, against a requirement of 0.5, and BBN would need mu_K >= 1.84e8 h/Mpc, 1.24e7 times BS's cap. BS25's DBI form fixes BBN (about 4e-6 to 7e-6), but on the spectra themselves v3 inside the galaxy cap stays far from Planck: best-fit chi2_plik 14,323 at the cap (not converged), 937 to 1,319 for the DBI sets, and it only approaches the bin count near mu_K = 2542 h/Mpc (598), which galaxies forbid. The board note quotes Delta chi2 against a LCDM refit; that comparison is not drawn here by the owner rule. Inherited inputs: the fits start from C7's LCDM refit and use the C2 likelihood's tau prior (0.0510 +- 0.0085); a0 does not enter at linear order, so the footings do not enter; omega_K is FITTED. The cap lambda_D = 1 DBI fit did not converge (plik 6,023, low-l TT infinite) and is not drawn in the right panel.

Sources: rebuild_2026_10_22/Q6_v3_class/q6_native_results.json @ ed5894d6e1;
rebuild_2026_10_22/Q6_v3_class/q6_native.py @ ed5894d6e1;
rebuild_2026_10_22/Q7_bs25_cosmology/q7_native_results.json @ 142d9d8ccd

Lanes: Q6 / Q7

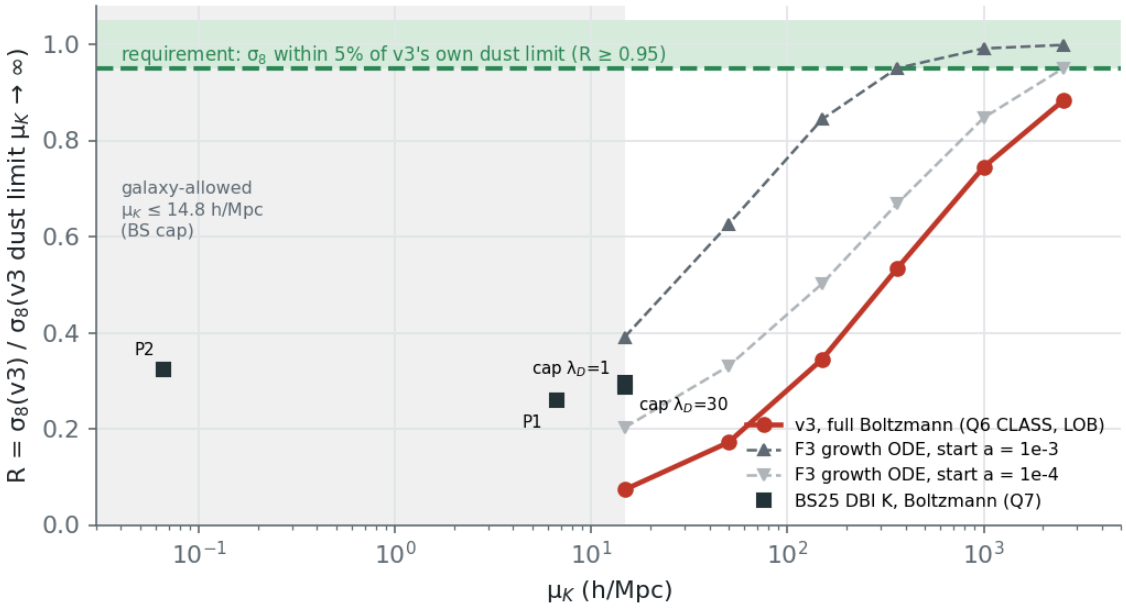
Linear growth, v3 Khronon

FAILS

On the board: Boltzmann σ8 0.061 at the cap (F3 ODE 0.391); 5% growth needs μ_K > 2542; cause is the BS cap (M1); Q7: BS25's DBI K fixes BBN only; growth R_B 0.26–0.32; BS ran no Boltzmann code

Linear growth, v3 Khronon

At the cap v3 keeps 7% of its own dust-limit σ8; even μ_K = 2542 h/Mpc misses the 95% line



source: rebuild_2026_10_22/Q6_v3_class/q6_native_results.json (stepC), rebuild_2026_10_22/Q7_bs25_cosmology/q7_native_results.json (stepC)

ON THE PLATE

Ratio R = sigma8(v3) / sigma8(v3's own dust limit mu_K -> infinity) at fixed parameters, from Q6's full Boltzmann runs (CLASS with the khronon patch, LOB model) at six mu_K, beside F3's growth ODE with the BSK2 hold kernel (start a = 1e-3 and 1e-4), and Q7's Boltzmann runs with BS25's DBI K.

HOW IT IS PLATED

x (log): mu_K in h/Mpc; grey shade: the galaxy-allowed range mu_K <= 14.8 h/Mpc (BS cap). y: R. Red: v3 Boltzmann; grey dashed: F3's ODE; black squares: DBI sets. Green dashed line and band: the framework's own requirement, R >= 0.95 (sigma8 within 5% of the dust limit).

WHY IT FAILS

At the BS cap v3 keeps R = 0.073 of its own dust-limit sigma8 (sigma8 0.061), lower than F3's ODE (0.391), and even at mu_K = 2542 h/Mpc R is 0.883, still below 0.95. The cause is v3's J'(0) = -1 hold, which switches Newtonian clustering off on scales k > a mu_K. The DBI K that fixes BBN does not help: R 0.26 to 0.32. The yardstick is v3's own dust limit, not LCDM; that limit is numerically identical to a CDM run, and its parameters are C7's LCDM refit, an inherited input. a0 does not enter at linear order, so both footings give the same numbers. The nonlinear escape is left to L1's run.

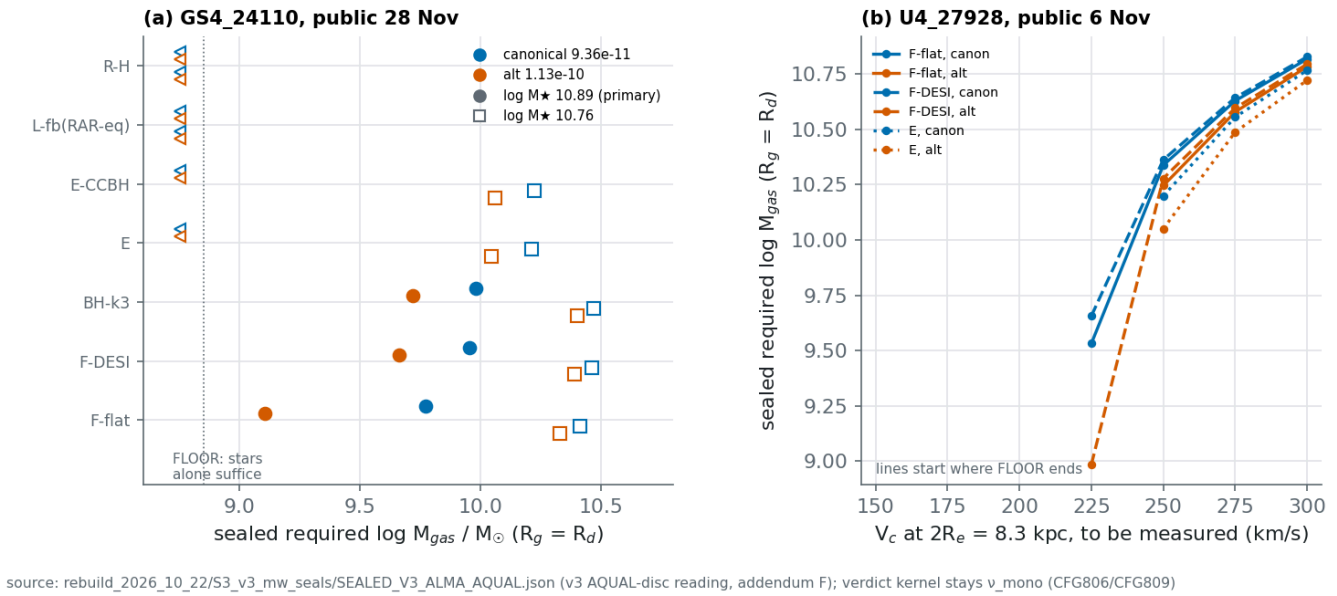
Sources: rebuild_2026_10_22/Q6_v3_class/q6_native_results.json @ ed5894d6e1;
rebuild_2026_10_22/Q7_bs25_cosmology/q7_native_results.json @ 142d9d8ccd;
rebuild_2026_10_22/F3_growth_audit/f3_growth_audit_results.json @ 059766caa5;
rebuild_2026_10_22/M1_trilemma/README.md @ 836cc3661f

Lanes: F3 / M1 / Q6 / Q7

ALMA U4 (6 Nov), GS4 (28 Nov)

OPEN

On the board: harness ready, expected unscorable; v3 kernel sealed as AG4e; addendum F filed (f6f0ba66a4): v3 AQUAL-disc reading; harness v3aq MOCK-tested



source: rebuild_2026_10_22/S3_v3_mw_seals/SEALED_V3_ALMA_AQUAL.json (v3 AQUAL-disc reading, addendum F); verdict kernel stays ν_mono (CFG806/CFG809)

ON THE PLATE

No data yet. The sealed v3 AQUAL-disc requirement (addendum F, a labelled parallel reading): for each stellar-mass branch and footing, the total gas mass v3 needs to match the sealed or measured rotation. GS4_24110: sealed V_c 206 km/s at R_e = 7.39 kpc, log M* 10.89 (primary) and 10.76. U4_27928: a grid in the V_c to be measured at 2R_e = 8.3 kpc.

HOW IT IS PLATED

Left: one row per branch; dots (log M* 10.89) and open squares (10.76) give the required log M_gas for R_g = R_d, blue canonical, orange alt; left-pointing triangles at the edge mean FLOOR (the stars alone already meet the dynamics). Right: required log M_gas against the U4 V_c still to be measured, for three branches and both footings; lines start where FLOOR ends.

WHY IT IS STILL OPEN

This tile is a prediction sealed before the data, not a result. On release day the plans' frozen rules compare the measured gas and rotation with these numbers, branch by branch and footing by footing. The verdict kernel stays nu_mono, and the v3 reading is scored beside it, never pooled. The harness is ready, but CFG942 expects a total-flux S/N well below 5 at native resolution, so the likeliest outcome on both dates is an upper limit. One disc excludes no model.

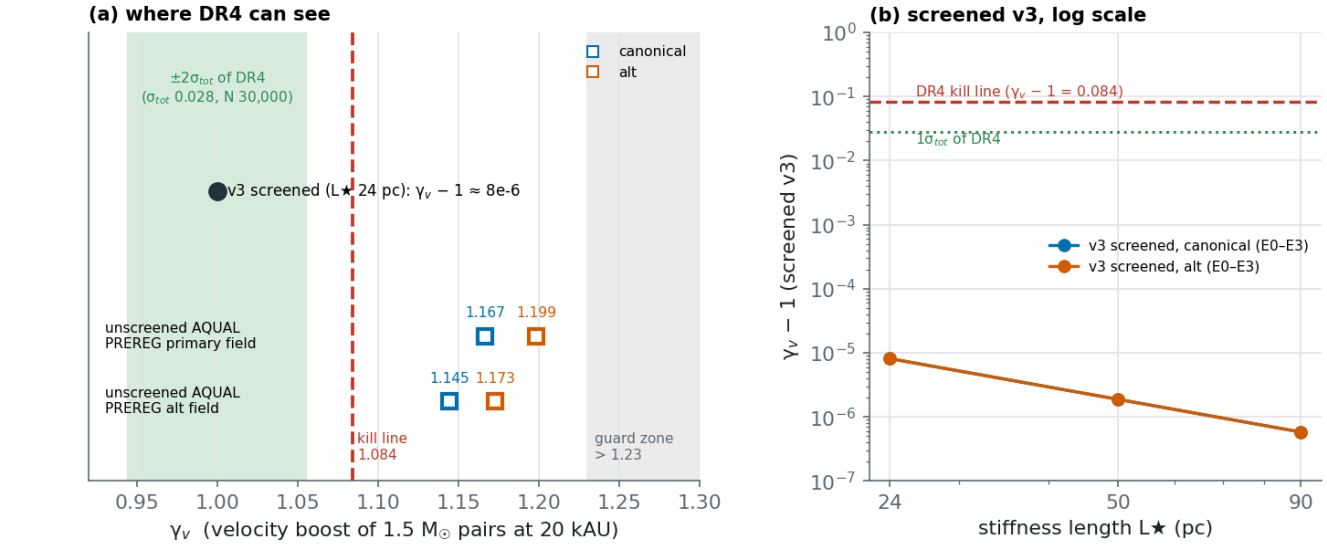
Sources: rebuild_2026_10_22/S3_v3_mw_seals/SEALED_V3_ALMA_AQUAL.json @ 90f4d517d6; campaign_fresh_gravity/CFG809_gs4_24110_ci21_scoring/CLARIFICATION_2026-10-23_ADDENDUM_F.md @ f6f0ba66a4; campaign_fresh_gravity/CFG942_alma_scoring_readiness/README.md @ a50a7fcb3f; argon_push/AG4e_n1_alma_seal/SEALED_N1.json @ b1bb9296db

Lanes: CFG806-U4 / CFG809 / CFG942 / D4 / AG4e

Gaia DR4 wide binaries (Dec)

OPEN

On the board: screened gamma-1 ~ 8e-6 vs AQUAL 1.167 / 1.199



source: covariant_push_2026_10_22/C6_literal_cassini/c6_literal_cassini_results.json (R3); doors_push_2026_10_22/E2_field_equation/e2_field_equation_results.json (R2); DR4 plan numbers: campaign_fresh_gravity/CFG943_dr4_readiness/README.md

ON THE PLATE

No data yet. Predictions for the velocity boost gamma_v of 1.5 Msun wide binaries at 20 kAU: v3 with its screening (C6: gamma_v - 1 at stiffness length L* = 24, 50, 90 pc, fields E0-E3, both footings) and the unscreened AQUAL contrast (E2, at the PREREG primary and alt external fields). The DR4 resolution and kill line come from the frozen DR4 plan as summarised in CFG943.

HOW IT IS PLATED

Left: gamma_v axis. Green band: +-2 sigma_tot of DR4 around 1 (sigma_tot 0.028 at N 30,000). Red dashed: the kill line 1.084. Grey: the guard zone above 1.23. Black dot: screened v3 (gamma_v - 1 about 8e-6, on top of 1). Open squares: unscreened AQUAL values, canonical blue and alt orange. Right (log): screened v3's gamma_v - 1 against L*, with the DR4 kill line and 1 sigma_tot drawn; canonical and alt lie on top of each other.

WHY IT IS STILL OPEN

Screened v3 predicts no wide-binary boost DR4 could see: gamma_v - 1 = 8.1e-6 at L* = 24 pc, falling to 5.8e-7 at 90 pc, about four orders of magnitude below DR4's sigma_tot. The unscreened AQUAL equation gives 1.167 (canonical) and 1.199 (alt) at the primary field, well above the kill line, so DR4 will tell the screened and unscreened readings apart if its systematics hold. CFG943 finds the plan READY in its code paths, but with a power gap against a MOND-size boost (44%, 2% and 59% kill rates in three mock classes).

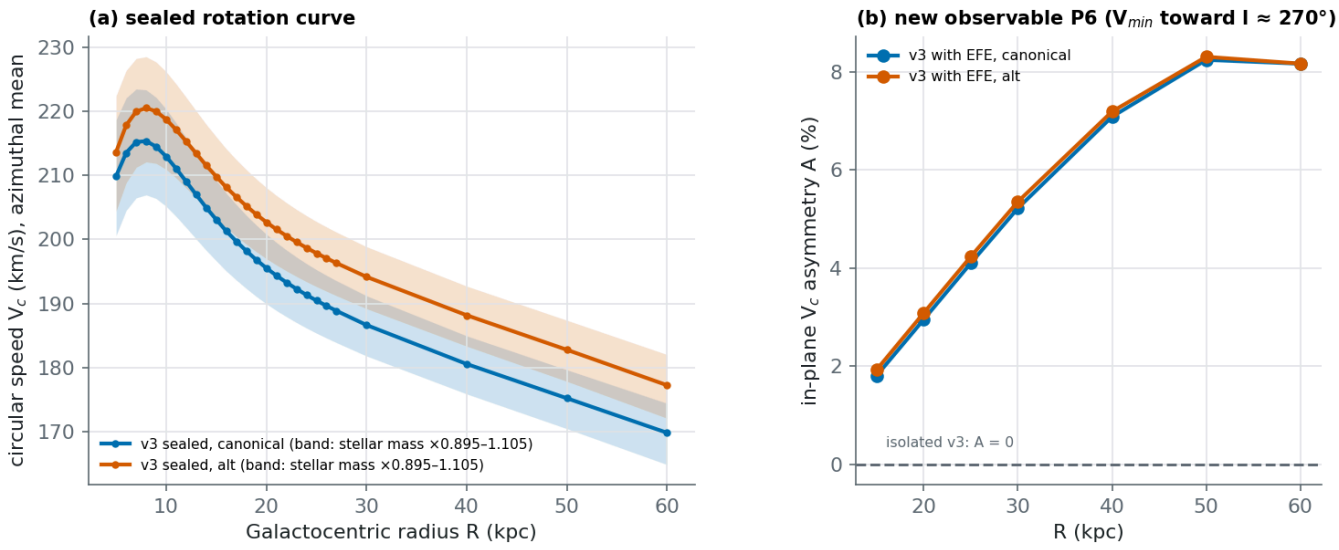
Sources: covariant_push_2026_10_22/C6_literal_cassini/c6_literal_cassini_results.json @ ae428dc252; doors_push_2026_10_22/E2_field_equation/e2_field_equation_results.json @ 474050bcd8; campaign_fresh_gravity/CFG943_dr4_readiness/README.md @ 7356bb671d

Lanes: C6 / E2 / CFG943

Gaia DR4 MW predictions

OPEN

On the board: v3 sealed: SEALED_V3_MW_DR4.json (15b7e6ff...); new observable: EFE-driven V_c asymmetry 3/5/8 % at 20/30/60 kpc; round-rule seal (10-10) kept



source: rebuild_2026_10_22/S3_v3_mw_seals/SEALED_V3_MW_DR4.json (sha256 15b7e6ff..., S3_v3_mw_seals); round-rule seal CFG553 kept, not drawn

ON THE PLATE

No DR4 data yet. The v3 sealed Milky Way predictions (S3): AQUAL disc with mu = x/(1+x), the external field solved, McMillan 2017 baryons with the stellar mass scaled by 0.895 to 1.105. Shown: the azimuthal-mean circular speed from 5 to 60 kpc and the new observable P6, the in-plane V_c asymmetry the external field creates.

HOW IT IS PLATED

Left: sealed V_c(R) for canonical (blue) and alt (orange), line at the census stellar mass, band over the stellar-mass scale. Right: P6 asymmetry A(R) in percent for both footings; the dashed zero line is what isolated v3 predicts.

WHY IT IS STILL OPEN

This is a sealed prediction, written before DR4 and kept apart from the old round-rule seal (CFG553, untouched and not drawn). The new handle is P6: with the external field, v3 predicts a V_c asymmetry of about 3% at 20 kpc, 5% at 30 kpc and 8% at 60 kpc, with V_min toward l = 270 degrees, while isolated v3 predicts none. The same lane finds v3's outer slope already SHAPE EXCLUDED against Gaia DR3 (a separate tile), so DR4 decides whether that failure stands. The NFW comparison band in the seal is not drawn.

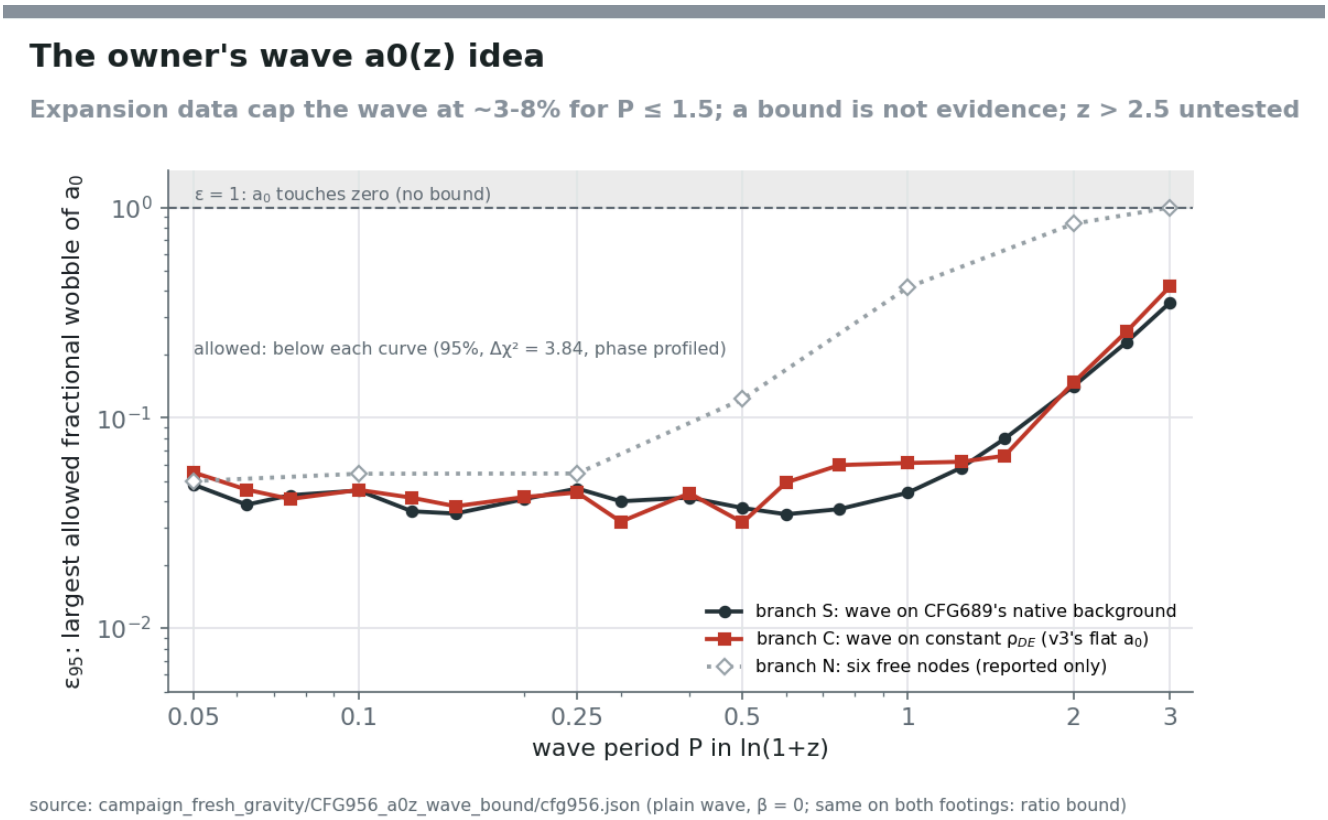
Sources: rebuild_2026_10_22/S3_v3_mw_seals/SEALED_V3_MW_DR4.json @ ec0c57d236; rebuild_2026_10_22/S3_v3_mw_seals/README.md @ a2e4a36598

Lanes: CFG553 / S3

The owner's wave a0(z) idea

OPEN

On the board: being tested, no result yet



source: campaign_fresh_gravity/CFG956_a0z_wave_bound/cfg956.json (plain wave, β = 0; same on both footings: ratio bound)

ON THE PLATE

The owner's idea (2026-10-21): a0(z) oscillates. CFG956 bounds a plain wave a0(z)/a0(0) = A(z)[1 + eps sin(2 pi ln(1+z)/P + phi)]/(1 + eps sin phi) with CFG689's own BAO + Pantheon+ likelihood, for 19 periods P from 0.05 to 3 in ln(1+z), on three backgrounds A(z).

HOW IT IS PLATED

x (log): wave period P in ln(1+z). y (log): eps_95, the largest wobble allowed at 95% (Delta chi2 = 3.84, phase profiled); anything below a curve is allowed. Black: wave on CFG689's native background; red: wave on constant rho_DE (v3's flat a0); grey dotted: six free nodes (reported only). Grey band above 1: a0 would touch zero, no bound.

WHY IT IS STILL OPEN

The expansion data allow a wave of at most about 3 to 8% in a0 for periods up to 1.5 in ln(1+z); longer periods act as a smooth tilt and loosen to 0.35 to 0.42 at P = 3. No branch prefers a wave (branch C has a local hint, Bonferroni not applied), and a bound on a wave is not evidence for one. Nothing above z of about 2.5 is constrained by expansion data. CFG957 finds no wave from the framework's own rates; one needs a new light scalar with two new constants. The bound is on the ratio, so it is the same on both footings. The tile stays open: the board note predates these bounds and the idea has no galaxy-side test yet.

Sources: campaign_fresh_gravity/CFG956_a0z_wave_bound/cfg956.json @ aaa000a458;
campaign_fresh_gravity/CFG956_a0z_wave_bound/README.md @ c929d7548a;
campaign_fresh_gravity/CFG957_a0z_wave_models_observables/README.md @ 12744bf082

Lanes: CFG956 / CFG957

~ Nightcaps ~

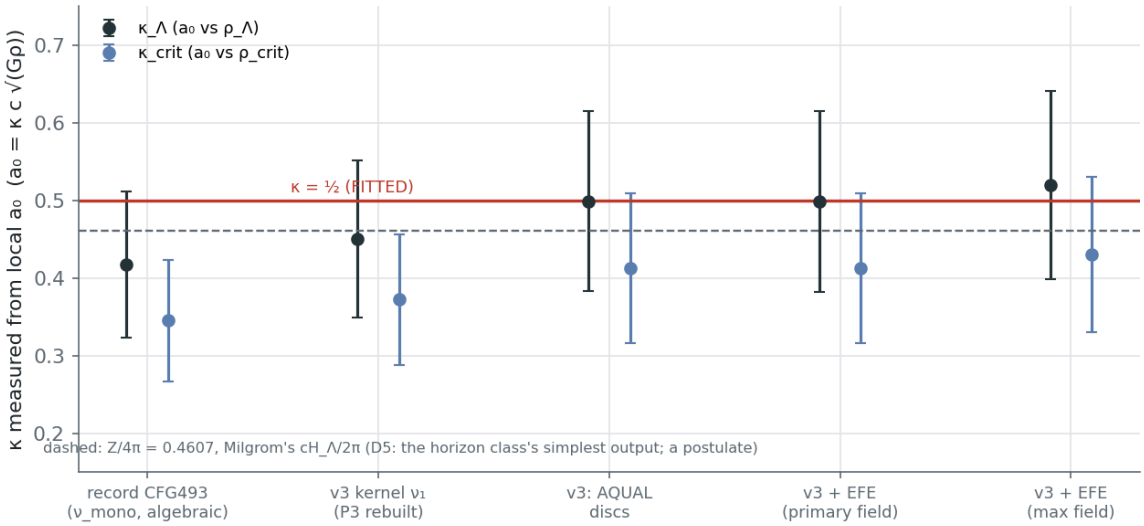
The deep why and the parent theory · 6 dishes

Why kappa = 1/2 ..... OPEN

On the board: fitted; no horizon construction gives it; S3 AQUAL discs: κ_Λ 0.499 ± 0.116, κ_crit 0.413 ± 0.096: consistent, not discriminating

Why kappa = 1/2

OPEN: κ_Λ 0.499 ± 0.116, κ_crit 0.413 ± 0.096: consistent with ½, not discriminating; no derivation



source: S3_v3_mw_seals/s3_partB_kappa_results.json; D5_kappa_four (Z, theorem T1–T3)

ON THE PLATE

κ in a₀ = κc√(Gρ), measured from the combined local a₀ of CFG493's three routes (P1 SPARC gas points, P2 MeerKAT MIGHTEE-HI widths, P3 WALLABY DR2), re-fitted by S3 step by step for v3 (kernel ν_1, AQUAL disc geometry, host field). κ_Λ uses ρ_Λ, κ_crit uses ρ_crit. D5 enumerates de Sitter horizon constructions.

HOW IT IS PLATED

κ_Λ (dark) and κ_crit (blue) with 1σ errors after each re-fit step. Red line: κ = ½, the fitted value the framework uses. Grey dashed line: Z/4π, which is Milgrom's cH_Λ/2π, the simplest value horizon thermodynamics gives (a postulate, not ½).

WHY IT IS STILL OPEN

With v3's AQUAL disc geometry, κ_Λ = 0.499 ± 0.116 and κ_crit = 0.413 ± 0.096: both are consistent with ½ but the error (about 0.1) cannot tell ½ from Z/4π or neighbouring values, so the data do not discriminate. On the theory side, D5 proves that products of horizon quantities give κ^2 = rational × odd power of π, so ½ is never reached; with square roots ½ appears only as 'κ = 1 times a half picked from a menu'. κ = ½ is therefore FITTED, and why it is ½ stays open.

Sources: rebuild_2026_10_22/S3_v3_mw_seals/s3_partB_kappa_results.json @ b5dc29d533; doors_push_2026_10_22/D5_kappa_four/d5_kappa_four_results.json @ 962007715e; rebuild_2026_10_22/S3_v3_mw_seals/README.md @ a2e4a36598; doors_push_2026_10_22/D5_kappa_four/README.md @ 962007715e

Lanes: CFG493 / CFG894 / D5 / S3

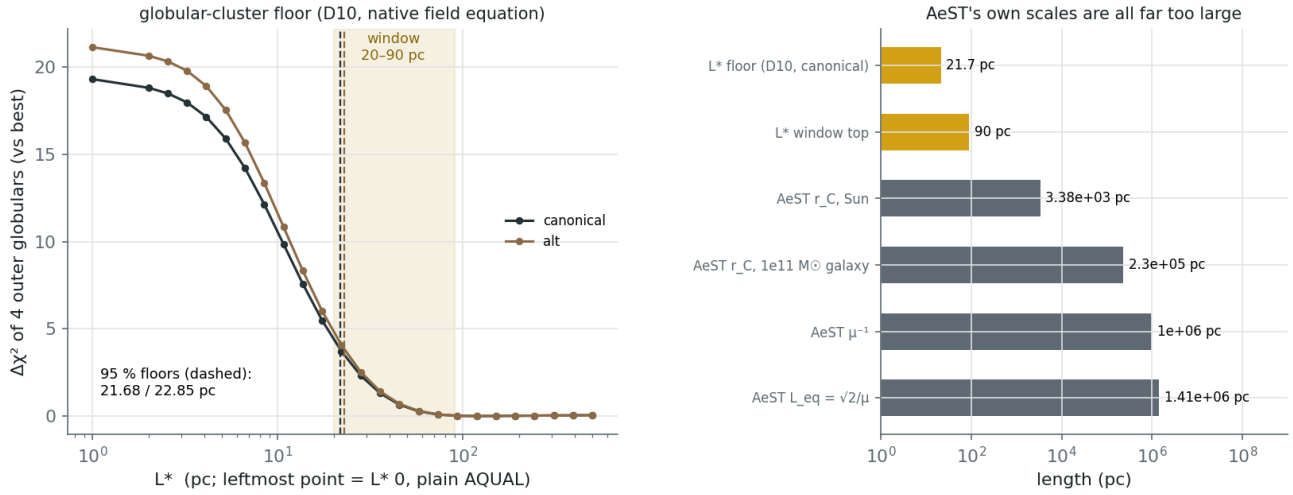
Stiffness length L*

OPEN

On the board: new fitted constant; floor ~22 pc, edge ~90 pc

Stiffness length L*

OPEN: a new FITTED constant; floor 21.7 / 22.8 pc, edge ~90 pc; no scale of AeST's class produces it (C3)



source: D10_native_Lstar/d10_native_lstar_results.json; C3_scale_map/c3_scale_map_results.json

ON THE PLATE

D10: velocity dispersions of four outer Milky Way globulars (NGC 2419, Pal 3, Pal 4, Pal 14), each solved as a Plummer sphere in the Galaxy's field with v3's stiffened field equation, scanned in L*. C3: the length scales of AeST's own action (mass term μ, r_C, curl length).

HOW IT IS PLATED

Left: $\Delta\chi^2$ of the globulars against L* for both footings; the leftmost point is L* = 0 (plain AQUAL); dashed lines mark the 95 % floor; the yellow band is the 20–90 pc window. Right: log bar chart of the floor and window top against AeST's own length scales (canonical).

WHY IT IS STILL OPEN

Plain AQUAL (L* = 0) is excluded by the globulars and no finite L* beats Newton, so the data give only a lower bound: L* ≥ 21.68 / 22.85 pc with SPS mass-to-light ratios (2.9–23.8 pc across the M/L and MW-field brackets). The upper edge near 90 pc comes from the window used in the lanes, not from a fit here. C3 shows no term in AeST's class makes such a short length: its scales start at kiloparsecs and its equivalent length √2/μ is ≥ 1.41 Mpc. So L* is a new constant, FITTED, and its origin is open.

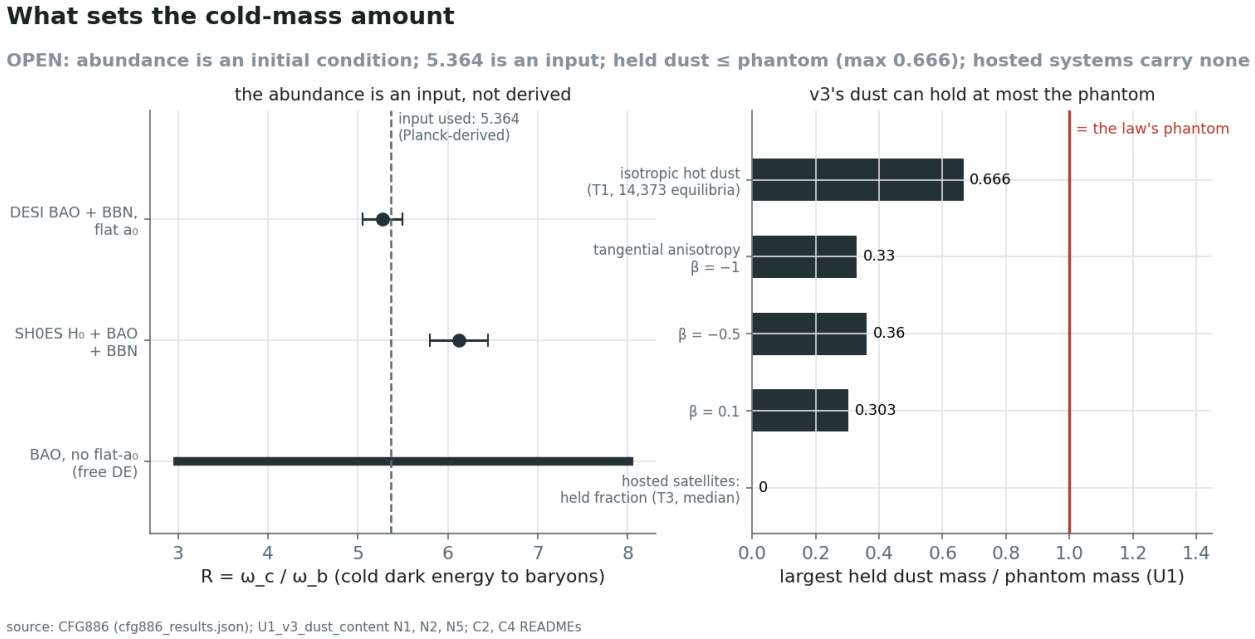
Sources: doors_push_2026_10_22/D10_native_Lstar/d10_native_lstar_results.json @ 3ac0573063;
covariant_push_2026_10_22/C3_scale_map/c3_scale_map_results.json @ fd7196584c;
doors_push_2026_10_22/D10_native_Lstar/README.md @ 3ac0573063;
covariant_push_2026_10_22/C3_scale_map/README.md @ fd7196584c

Lanes: C3 / D10

What sets the cold-mass amount

OPEN

On the board: dust abundance is an initial condition; 5.364 an input; U1: held dust ≤ phantom (theorem T1); hosted systems carry none (T3)



source: CFG886 (cfg886_results.json); U1_v3_dust_content N1, N2, N5; C2, C4 READMEs

ON THE PLATE

CFG886 rebuilds the cold-to-baryon ratio R = ω_c/ω_b from DESI DR2 BAO + BBN deuterium (and a SH0ES H0 variant, and a free-dark-energy scan) without the Planck fit. U1 solves v3's dust equilibria for 137 system-footing cases (satellites, Coma UDGs, TDGs, DF2/DF4) and asks how much dust v3 can hold.

HOW IT IS PLATED

Left: R from the three CFG886 routes (points with 1σ, the free-DE range as a bar) against the 5.364 input the lanes use (dashed). Right: U1's largest held-dust-to-phantom mass ratio for isotropic and anisotropic dust, and the median held fraction for hosted satellites; the red line at 1 is the law's phantom.

WHY IT IS STILL OPEN

In v3 (C2, C4) the dust abundance is a free initial condition, so nothing in the theory sets it; the value 5.364 is an input taken from Planck's ΛCDM fit (inherited input). CFG886 recovers 5.27 ± 0.22 only by assuming constant dark energy, gets 6.12 ± 0.32 with the local H0, and with free dark energy BAO + BBN allow 2.95–8.06: not pinned. Locally, U1's theorem T1 says held dust never exceeds the phantom (largest ratio 0.666 over 14,373 equilibria; radial anisotropy β > ¼ can exceed it only near the centre, not shown), and T3 says hosted systems hold none. The amount is not derived anywhere.

Sources: campaign_fresh_gravity/CFG886_ratio_fret_without_planck/cfg886_results.json @ 5614bd0a57; rebuild_2026_10_22/U1_v3_dust_content/u1_v3_dust_content_results.json @ bbb6bfcc71; campaign_fresh_gravity/CFG886_ratio_fret_without_planck/README.md @ 5614bd0a57; rebuild_2026_10_22/U1_v3_dust_content/README.md @ 58de634d74; covariant_push_2026_10_22/C2_aest_cosmology/README.md @ 5e93769089; covariant_push_2026_10_22/C4_khronon_parent/README.md @ 5f4f845554

Lanes: C2 / C4 / CFG886 / U1

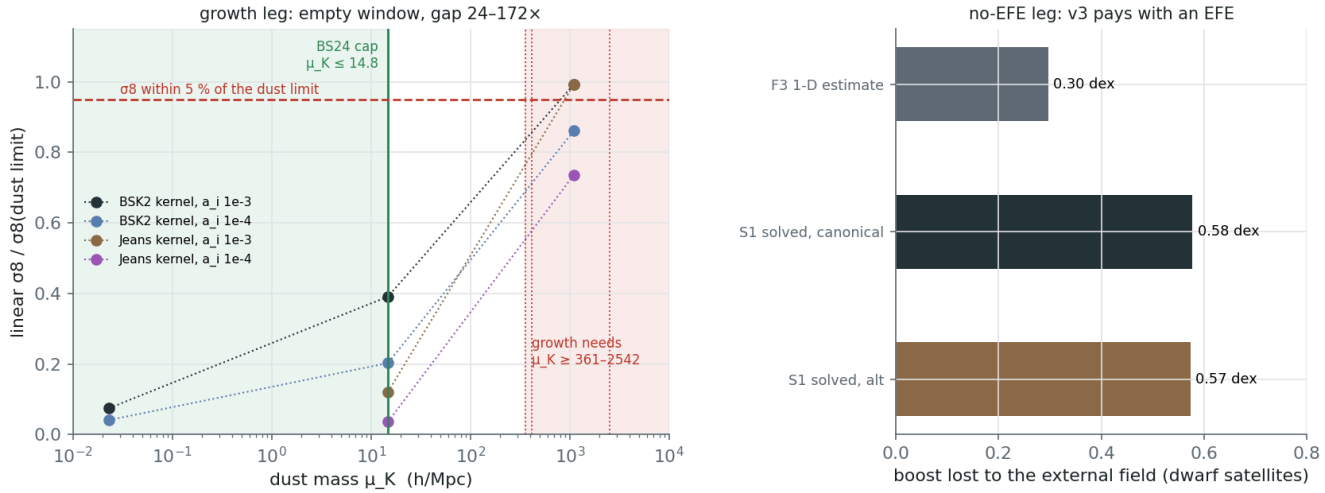
Law + a^1 growth + no EFE, local

FAILS

On the board: no-go stands; v3 pays the no-EFE leg with an EFE, and its linear growth fails (F3)

Law + a¹ growth + no EFE, local

FAIL: the no-go stands; v3 evades 'no EFE' by having one, and its growth leg has no allowed μ_K



source: F3_growth_audit/f3_growth_audit_results.json (K4, K5, K6); S1_v3_hosted_small; CFG946 README

ON THE PLATE

A theory statement (CFG946 theorem: a local second-order field that gives the profile-independent deep law must carry an external-field effect), checked for v3 by F3: linear σ8 of v3's Khronon dust as a function of its mass μ_K, for two kernels and two start times, and the EFE cost on dwarf satellites (F3's 1-D estimate, S1's solved value).

HOW IT IS PLATED

Left: linear σ8 relative to the dust limit at three μ_K values per kernel (log axis). Green: μ_K allowed by BS24's Eq. 39 cap; red: μ_K needed for σ8 within 5 % (dashed line). The two never meet. Right: the boost the satellites lose to the external field (F3 estimate vs S1's solve).

WHY IT FAILS

The no-go says no local theory can have the law, standard a^1 growth and no EFE at once. v3 does not dodge it: it pays the no-EFE leg by having an EFE (which costs the satellites 0.58 / 0.57 dex, about twice F3's 1-D estimate), and the growth leg comes back as an empty μ_K window: growth needs μ_K ≥ 361–2542 h/Mpc, the cap is 14.8, a gap of 24–172×. Inherited inputs: F3's linear runs use Planck-2018 initial conditions and a flat w = −1 background (ΛCDM-processed).

Sources: rebuild_2026_10_22/F3_growth_audit/f3_growth_audit_results.json @ 059766caa5; rebuild_2026_10_22/S1_v3_hosted_small/s1_v3_hosted_small_results.json @ 2b6bb658a1; rebuild_2026_10_22/F3_growth_audit/README.md @ a77915b2de; campaign_fresh_gravity/CFG946_option2_phantom_not_cold/README.md @ 124a3add65

Lanes: CFG941 / CFG944 / CFG946 / CFG947 / F3

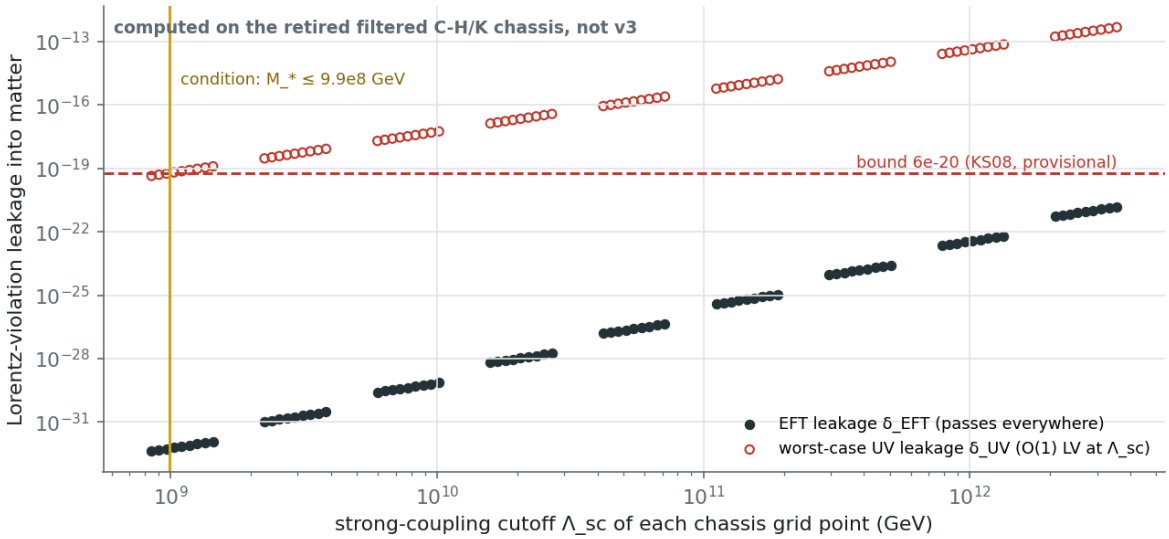
Radiative stability (G12)

NOT YET RECOMPUTED

On the board: was conditional: needs UV new physics < ~1e9 GeV (filtered chassis)

Radiative stability (G12)

NOT YET RECOMPUTED FOR v3 (was conditional: UV new physics needed at ≤ 9.9e8 GeV)



source: CFG320_radiative_stability_g12/cfg320_radiative_stability_results.json (old chassis); N0_board_audit

ON THE PLATE

CFG320's gate G12 on the old filtered C-H/K chassis: an 81-point grid of the couplings a_c and c_2, each with its own strong-coupling cutoff Λ_{sc}, and the Lorentz-violation leakage into matter from loops (EFT piece) and from a worst-case UV completion, against a 2σ bound from Klinkhamer & Schreck 2008 (marked provisional in the lane). The board tile is grey: not yet recomputed for v3 (N0 audit row 54).

HOW IT IS PLATED

Log-log scatter over the 81 grid points: EFT leakage (dark dots) and worst-case UV leakage (open red) against each point's cutoff, the bound as a dashed line, and the condition $M_* \le 9.9\text{e}8$ GeV as a vertical line.

WHY IT IS STILL OPEN

On the old chassis the theory is natural (loop corrections to the couplings stay tiny) and the EFT leakage is at least 40 times under the bound at every point, but a worst-case UV completion passes at only 3 of 81 points. So the old verdict was conditional: new UV physics must enter below about 1e9 GeV. v3 also has a preferred foliation, so this bound must be redone with v3's own couplings (batch R5); until then the tile is grey and the old numbers are shown only as the record.

Sources: campaign_fresh_gravity/CFG320_radiative_stability_g12/cfg320_radiative_stability_results.json @ d96517c30a; campaign_fresh_gravity/CFG320_radiative_stability_g12/README.md @ d96517c30a; rebuild_2026_10_22/N0_board_audit/AUDIT.md @ 522babce9a

Lanes: CFG320 / N0

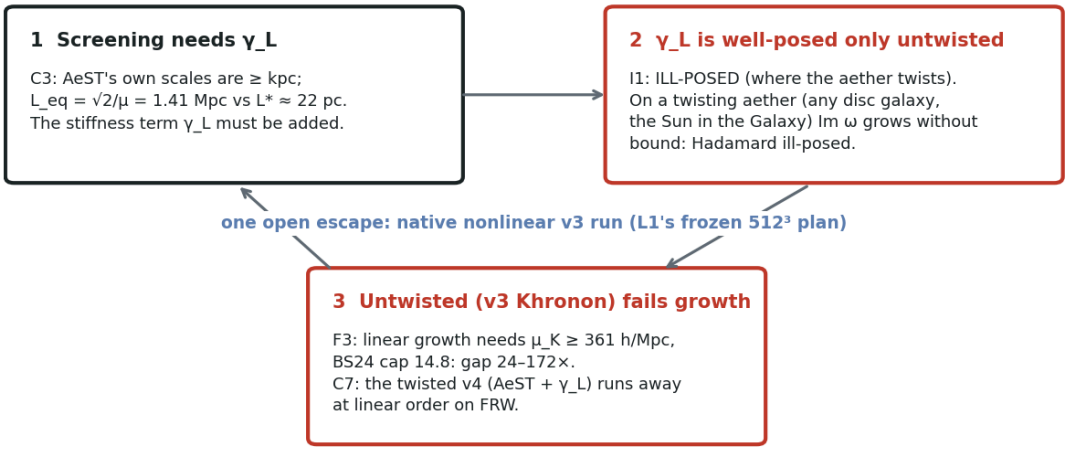
Covariant completion

FAILS

On the board: trilemma: screening needs γ_L; γ_L well-posed only untwisted; untwisted (v3) fails growth; open: native nonlinear v3 run

Covariant completion

FAIL: trilemma; screening needs γ_L, γ_L works only untwisted, untwisted (v3) fails growth



source: C3_scale_map, I1_gammaL_causality, F3_growth_audit (results json); C7_aest_class README

ON THE PLATE — No data arrays: a theory tile. Three lanes form the trilemma: C3 (AeST's own length scales), I1 (principal symbol and Cauchy problem of the γ_L stiffness term on twisting and untwisted aethers), F3 (linear growth of the untwisted v3 Khronon), with C7 (v4 = AeST + γ_L in a patched CLASS) for the twisted option on FRW.

HOW IT IS PLATED — Three boxes, one per horn, each with its deciding number, and arrows showing how fixing one horn forces the next; the centre names the one escape still open.

WHY IT FAILS — A covariant parent must (1) screen the law on small scales, which needs the added stiffness term γ_L because AeST's own scales are all far too large; (2) keep the field equations well-posed, but I1 shows γ_L is well-posed only where the aether is twist-free, and a disc galaxy or the Sun in the Galaxy forces twist; (3) the twist-free choice is v3's Khronon, whose linear growth fails (no allowed μ_K window), while the twisted v4 runs away on FRW (C7). Only a native nonlinear v3 simulation could still rescue growth; until it runs, the completion fails.

Sources: covariant_push_2026_10_22/C3_scale_map/c3_scale_map_results.json @ fd7196584c;
rebuild_2026_10_22/I1_gammaL_causality/i1_gammal_causality_results.json @ 81af28477c;
rebuild_2026_10_22/F3_growth_audit/f3_growth_audit_results.json @ 059766caa5;
doors_push_2026_10_22/D10_native_Lstar/d10_native_lstar_results.json @ 3ac0573063;
rebuild_2026_10_22/I1_gammaL_causality/README.md @ 81af28477c;
covariant_push_2026_10_22/C7_aest_class/README.md @ 64b0fdc9b4;
rebuild_2026_10_22/F3_growth_audit/README.md @ a77915b2de

Lanes: I1 / C7 / F3